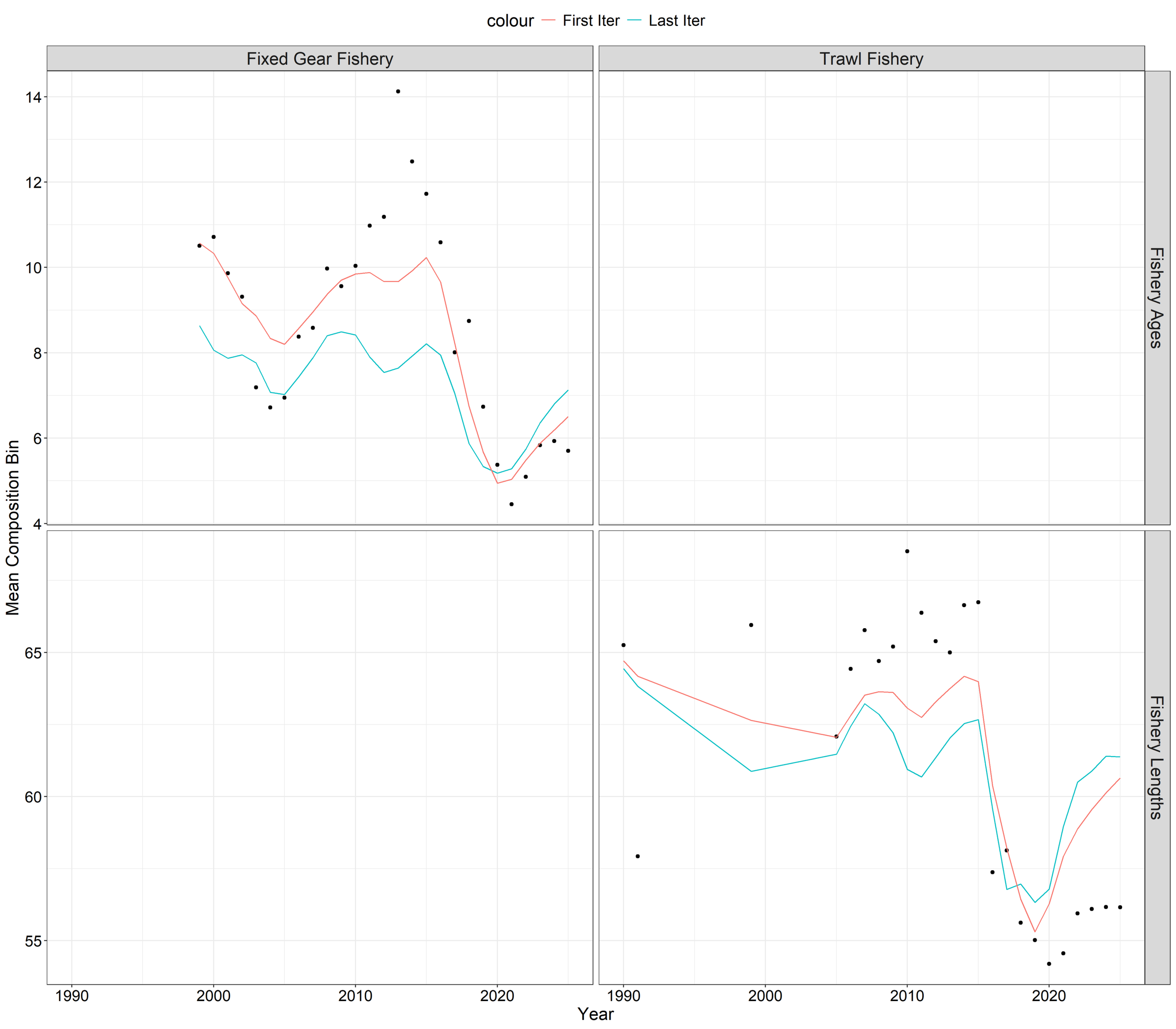
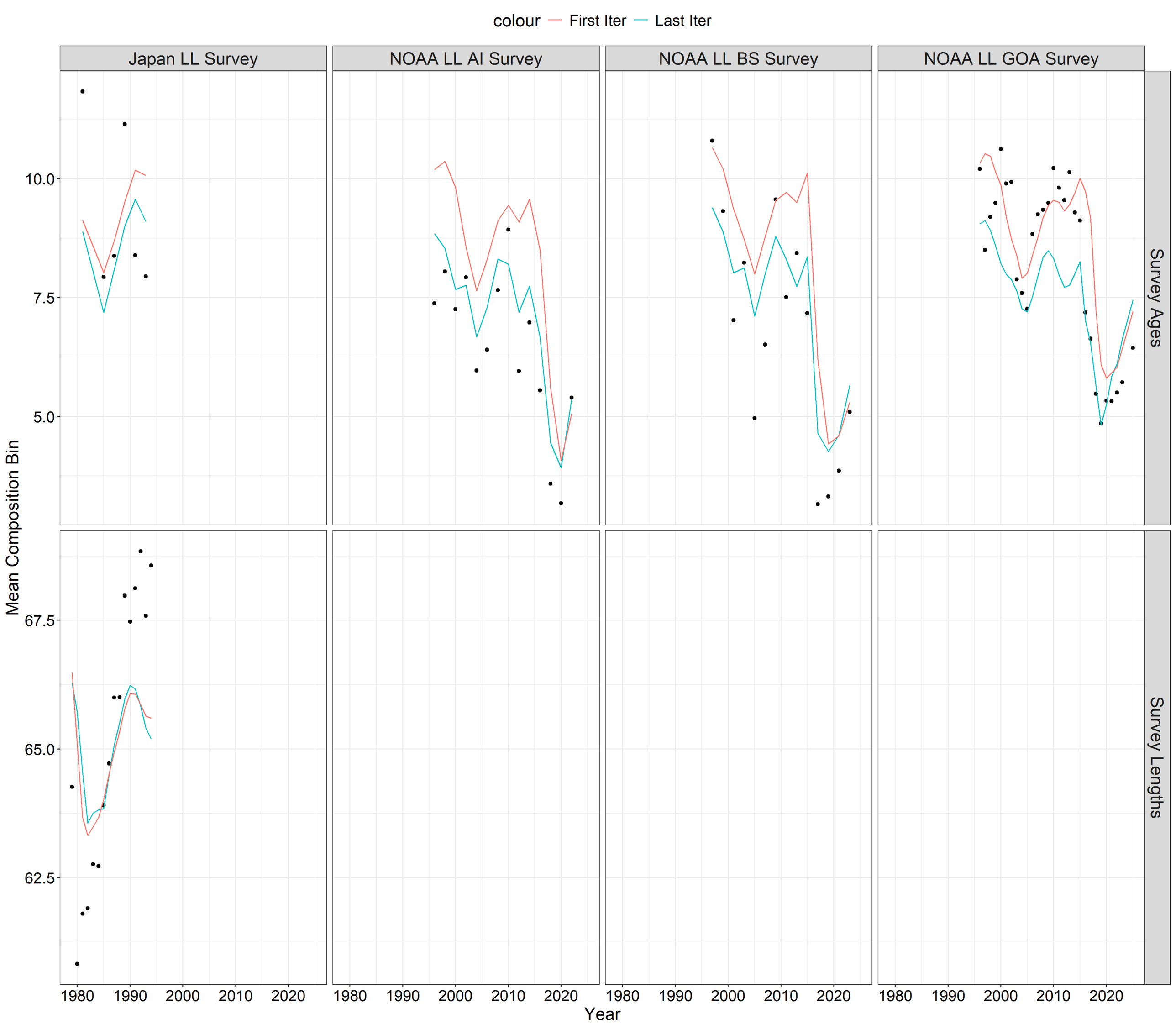
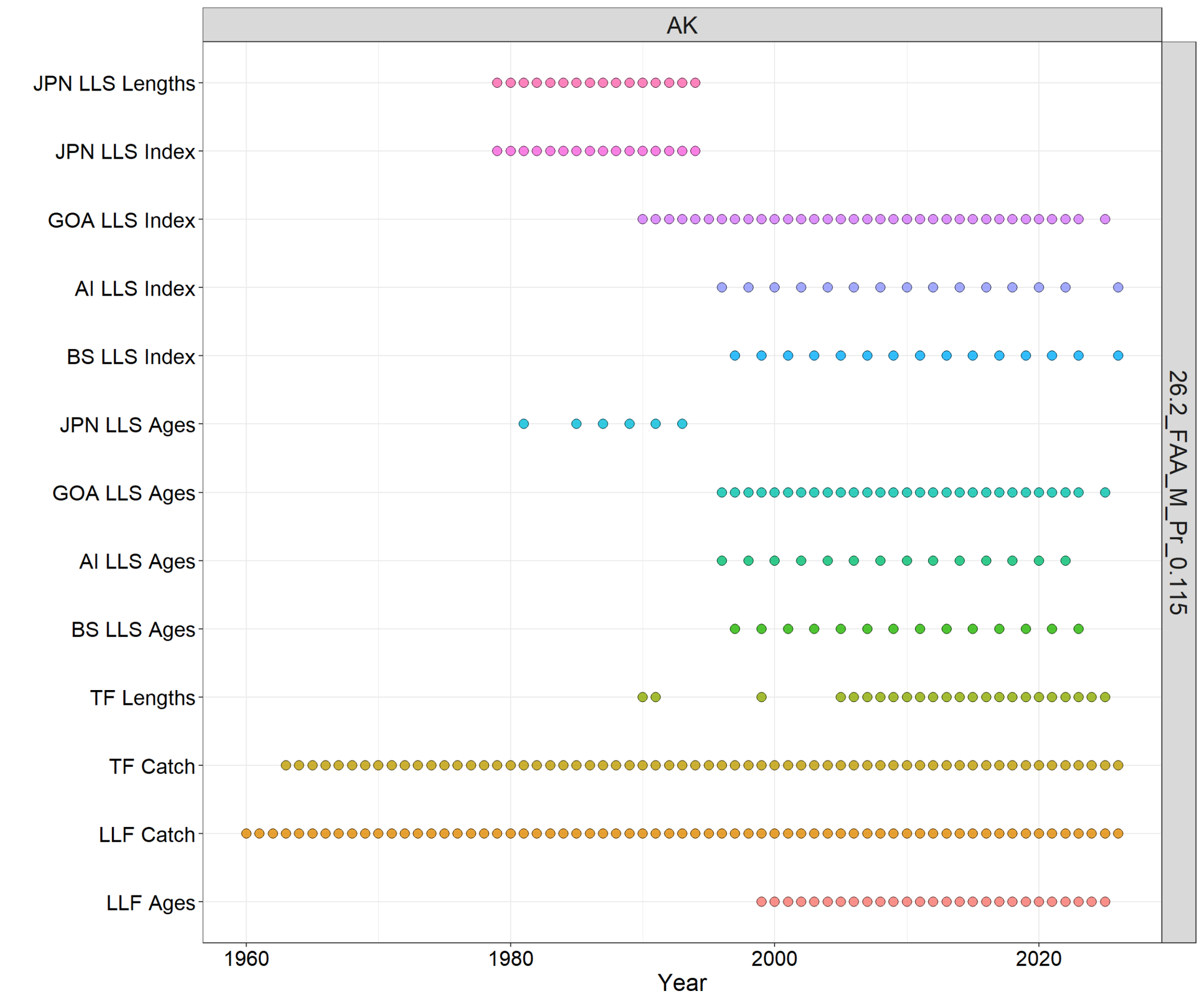
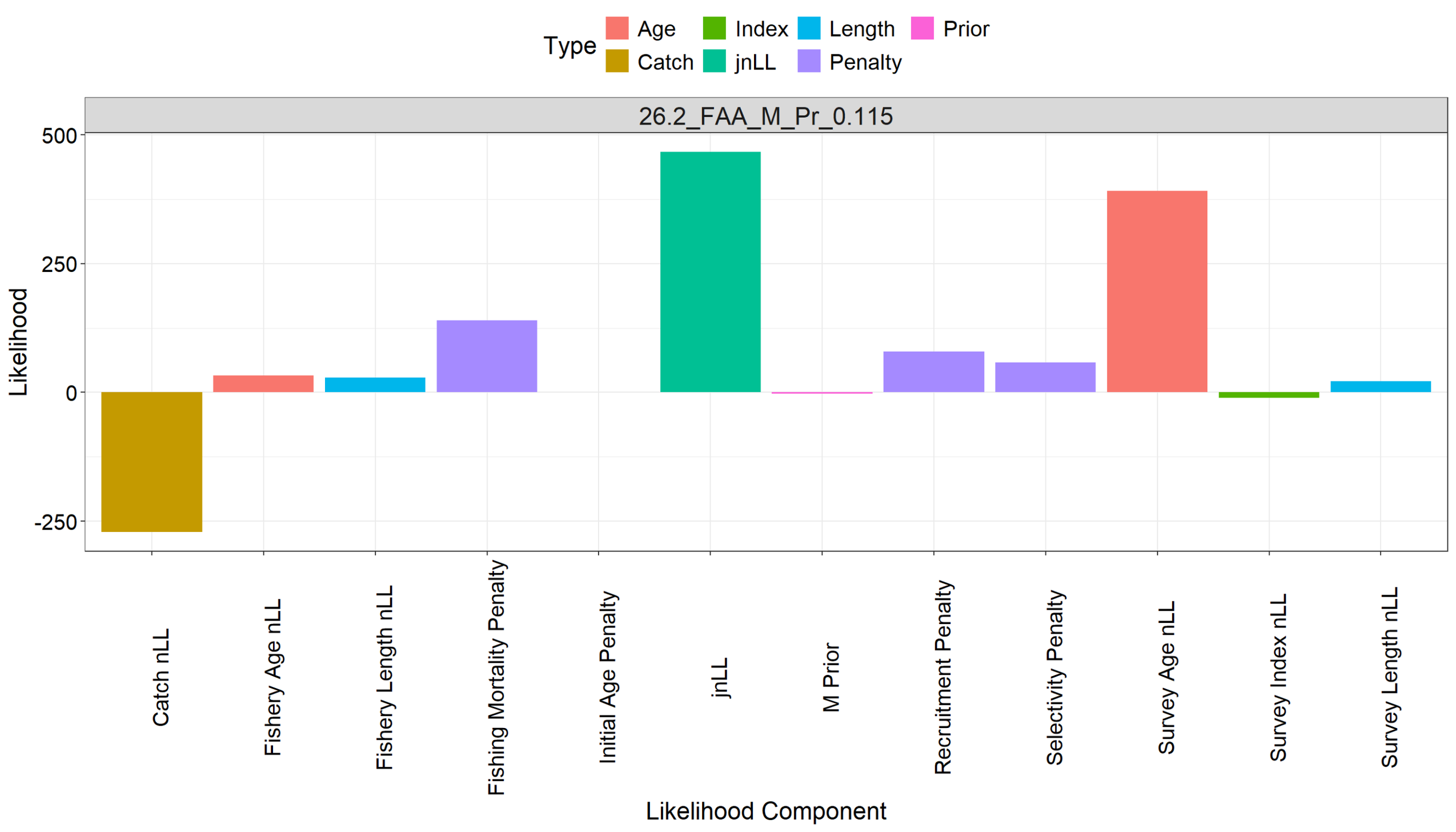

colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year


colour
First Iter
Last Iter
Japan LL Survey
NOAA LL AI Survey
NOAA LL BS Survey
NOAA LL GOA Survey
Survey Ages
Survey Lengths
Mean Composition Bin
10.0
7.5
5.0
67.5
65.0
62.5
1980
1990
2000
2010
2020
Year


AK
JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages
26.2_FAA_M_Pr_0.115
1960
1980
2000
2020
Year


Type
Age
Index
Length
Prior
Catch
jnLL
Penalty
26.2_FAA_M_Pr_0.115
500
250
0
-250
Likelihood
Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL
Likelihood Component

# Numbers-at-Age

Sex — Female ····· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

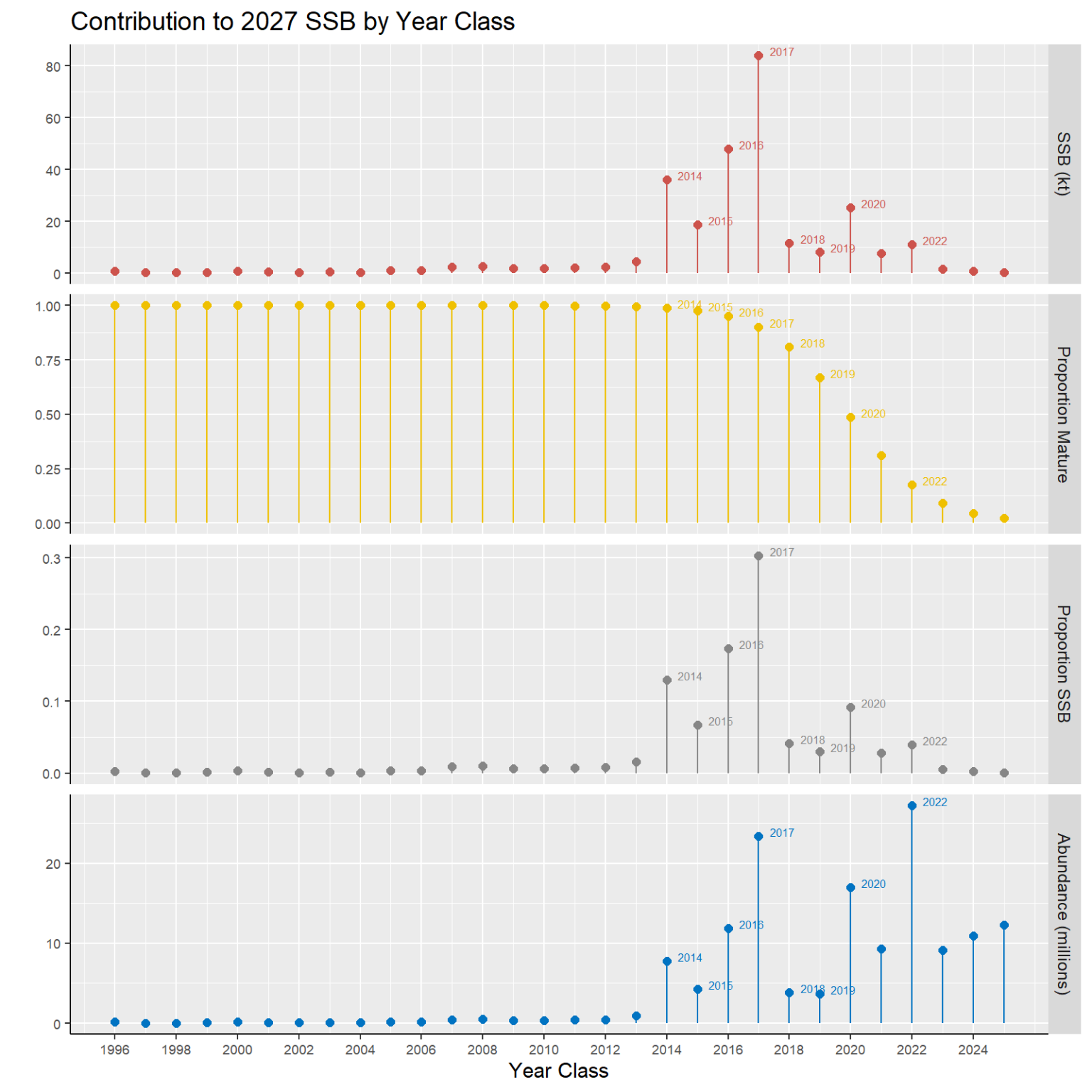
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

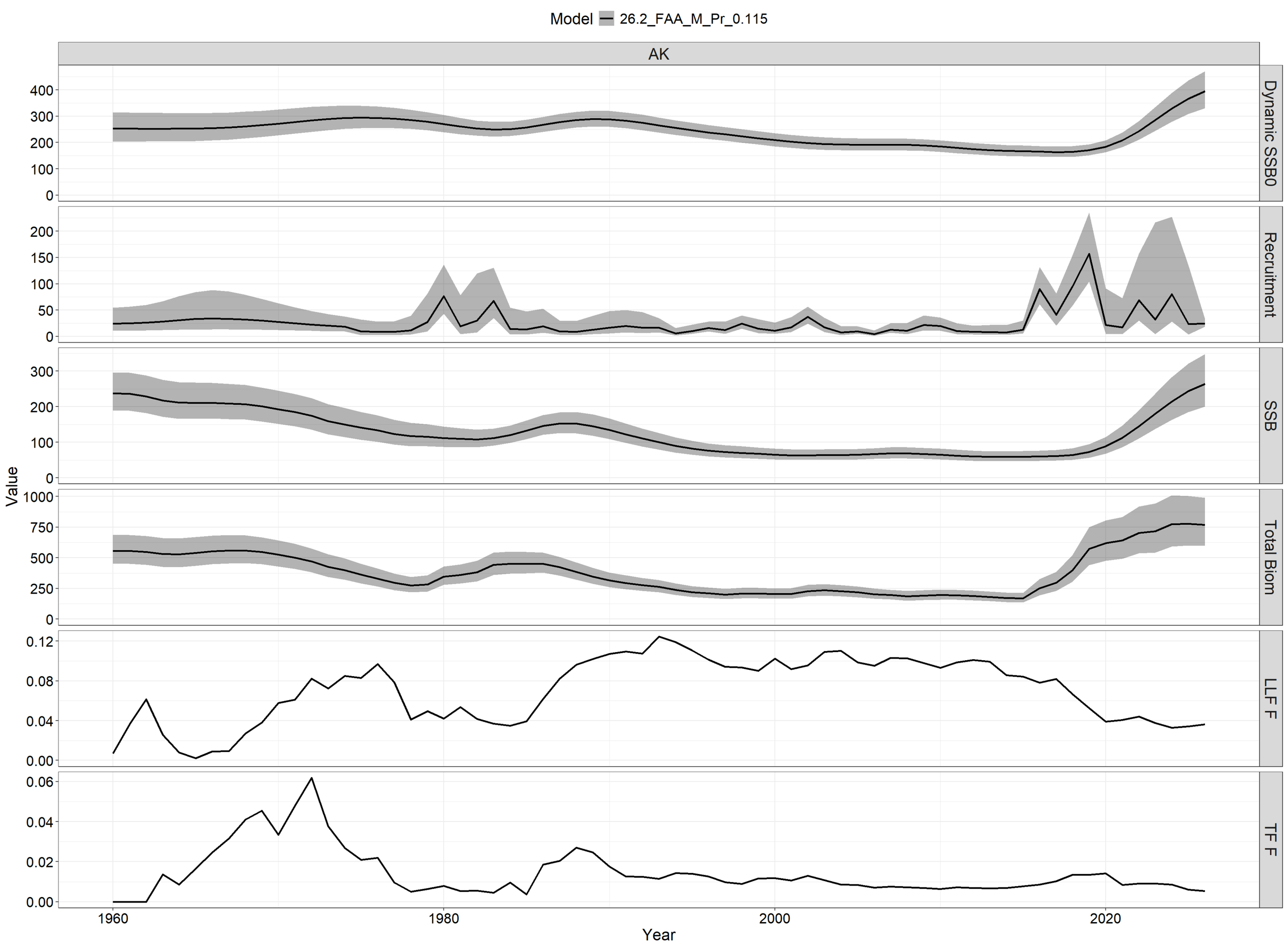
Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

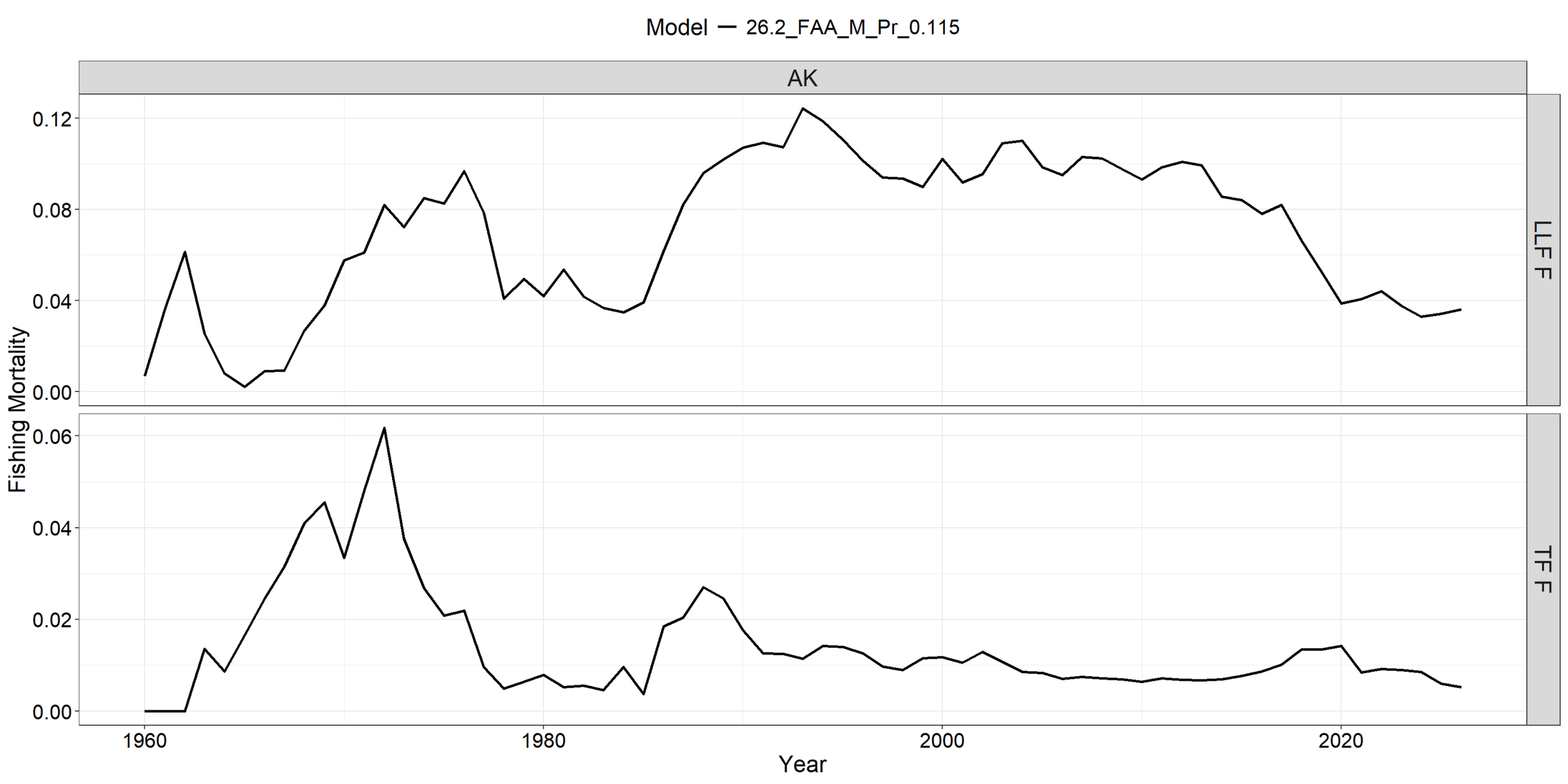
Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

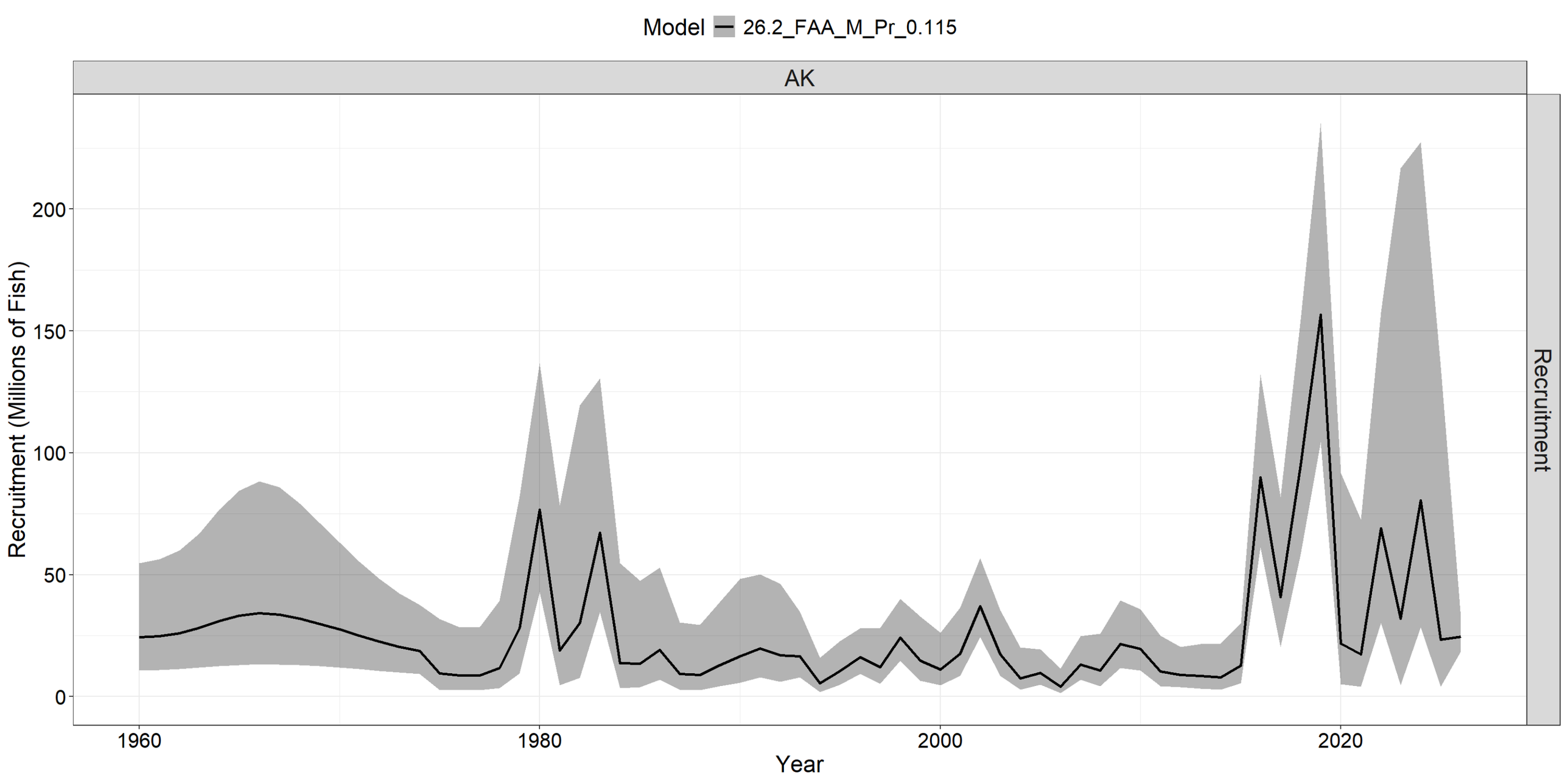
Numbers-at-Age (millions)

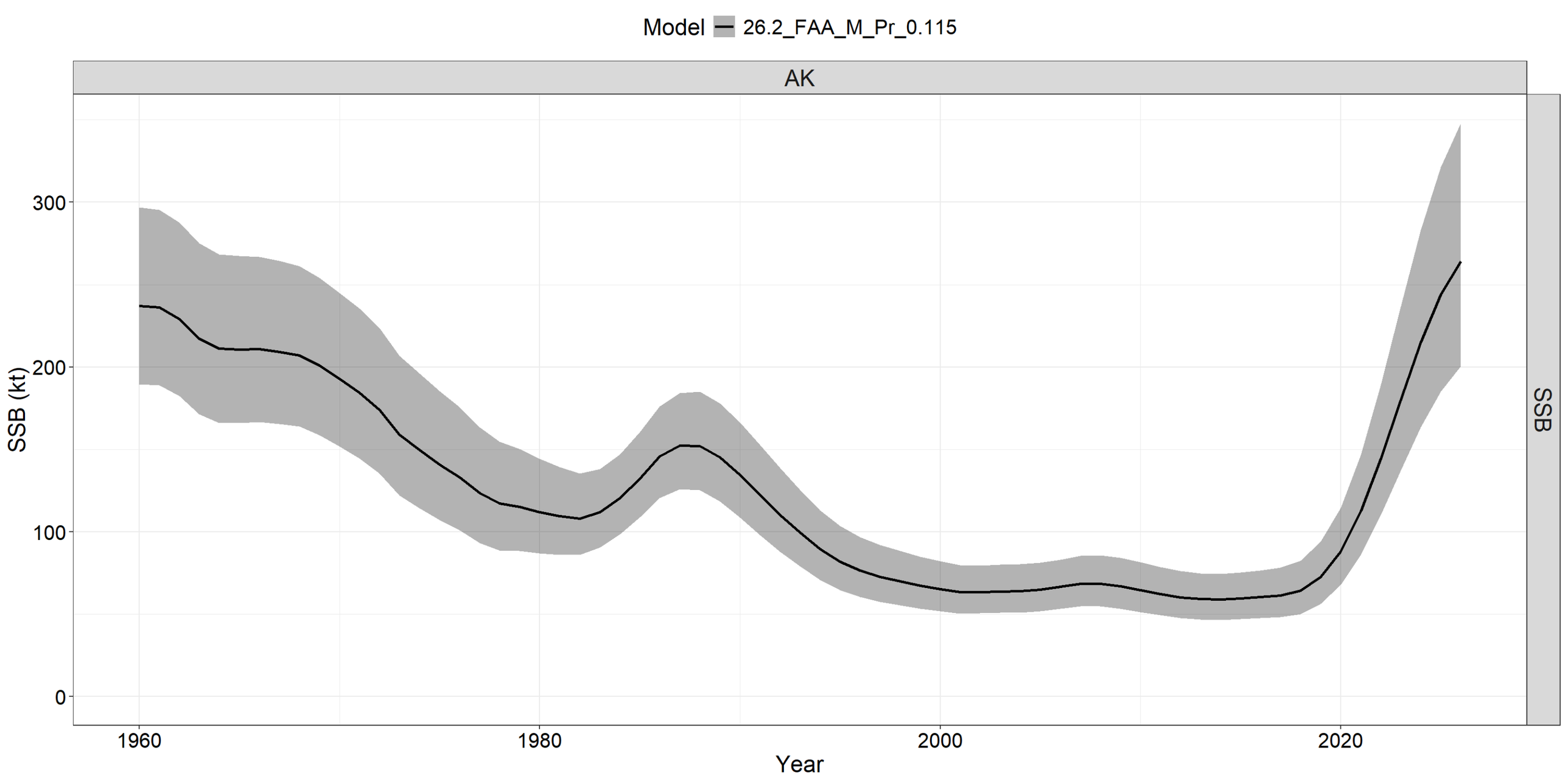
Year

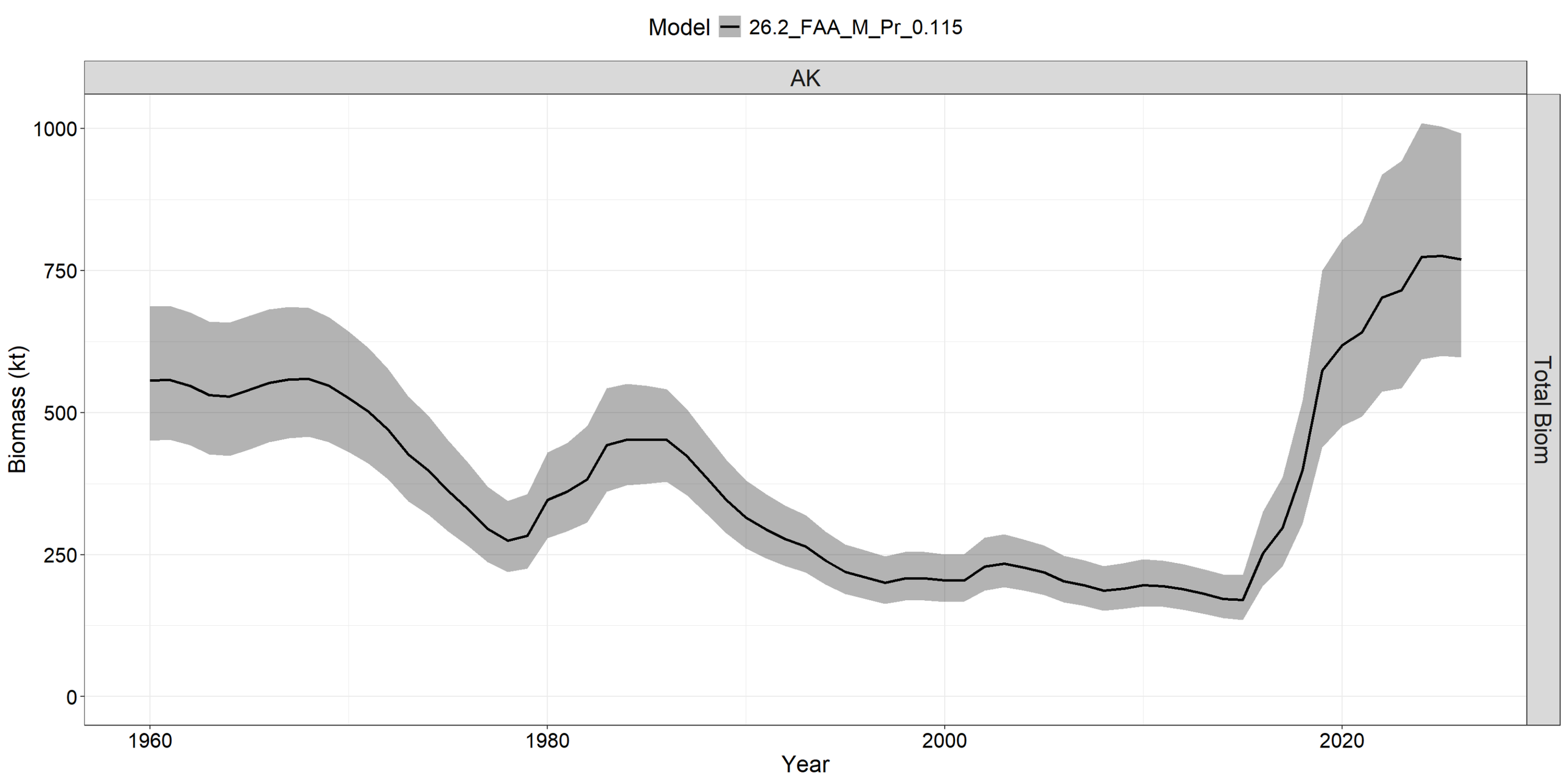

Contribution to 2027 SSB by Year Class
SSB (kt)
Proportion Mature
Proportion SSB
Abundance (millions)
Year Class
2017
2016
2014
2015
2020
2018
2019
2022
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024

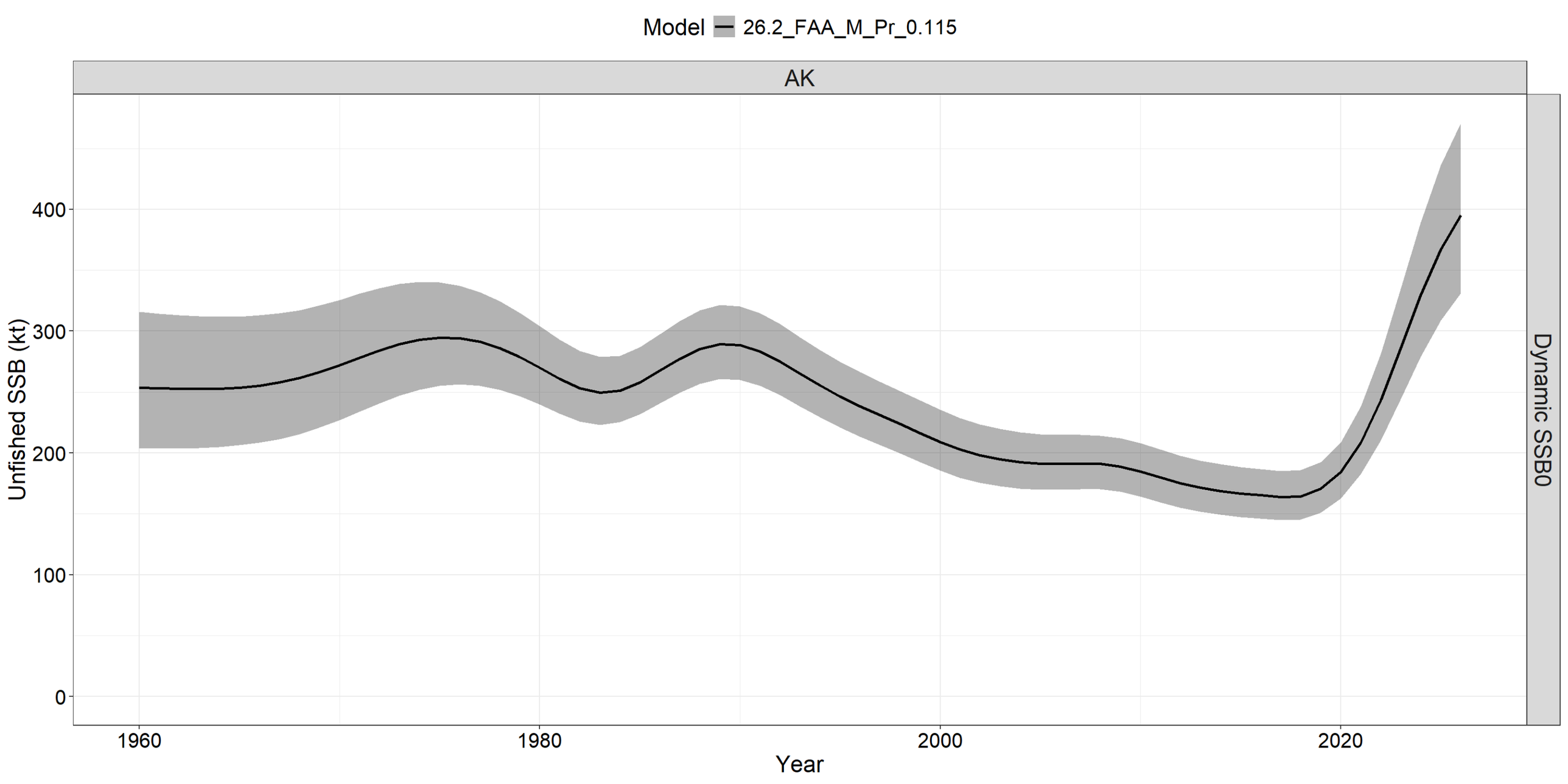

Model 26.2_FAA_M_Pr_0.115
AK
Dynamic SSB0
Recruitment
SSB
Total Biom
LLF F
TF F
Value
Year
1960
1980
2000
2020

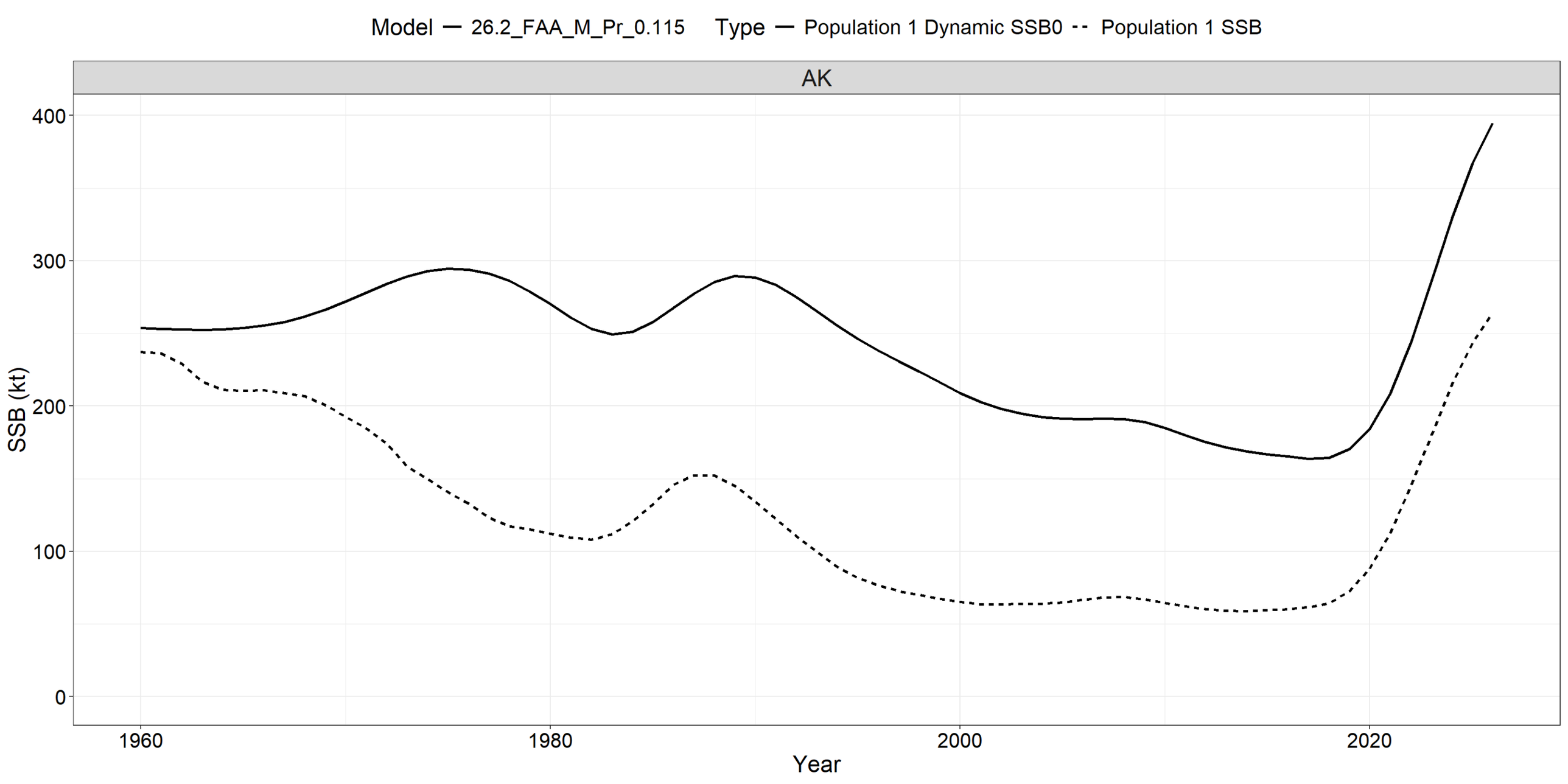

Model — 26.2_FAA_M_Pr_0.115
AK
LLF F
TF F
Fishing Mortality
0.12
0.08
0.04
0.00
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year

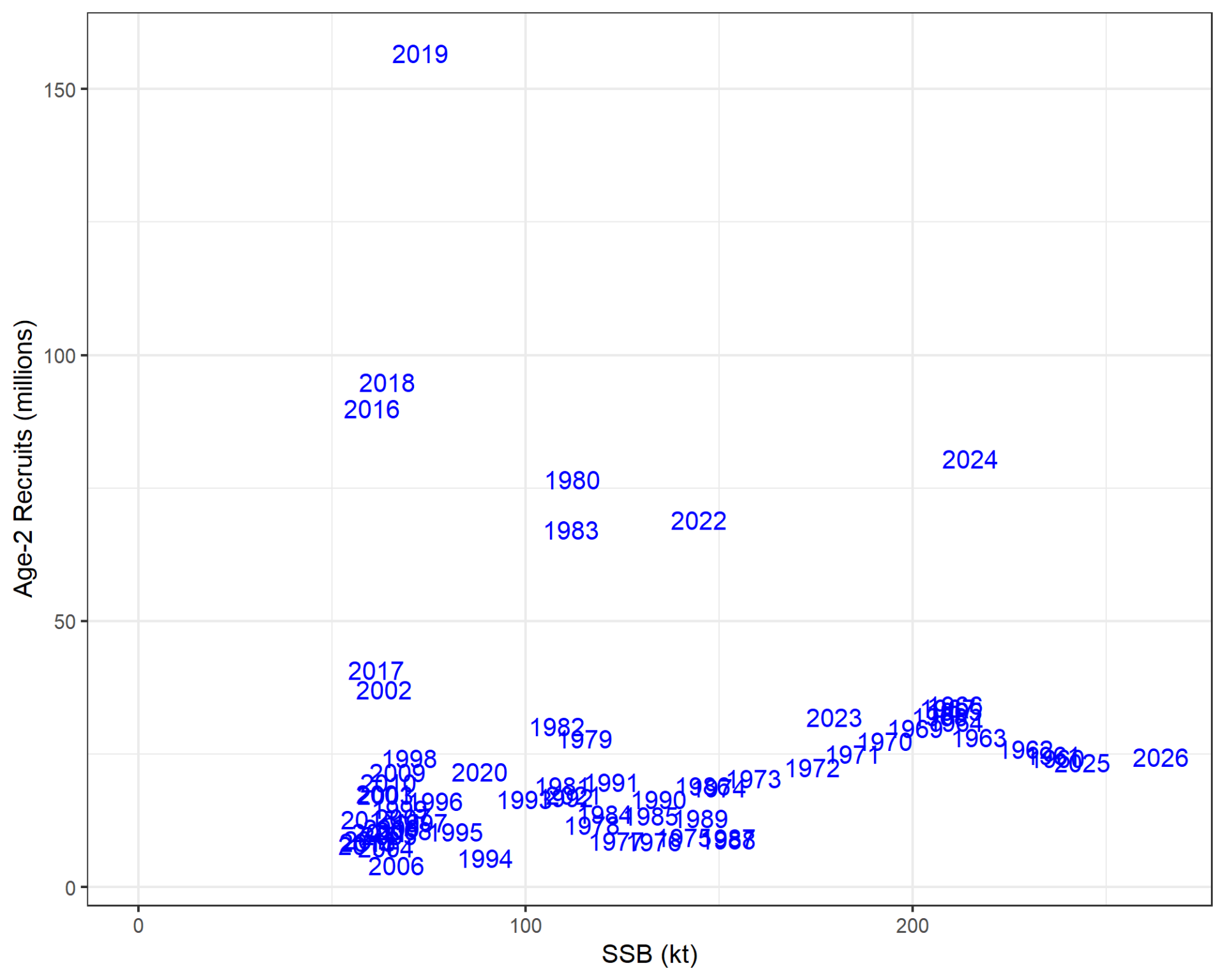

Model
26.2_FAA_M_Pr_0.115
AK
Recruitment (Millions of Fish)
0
50
100
150
200
1960
1980
2000
2020
Year
Recruitment

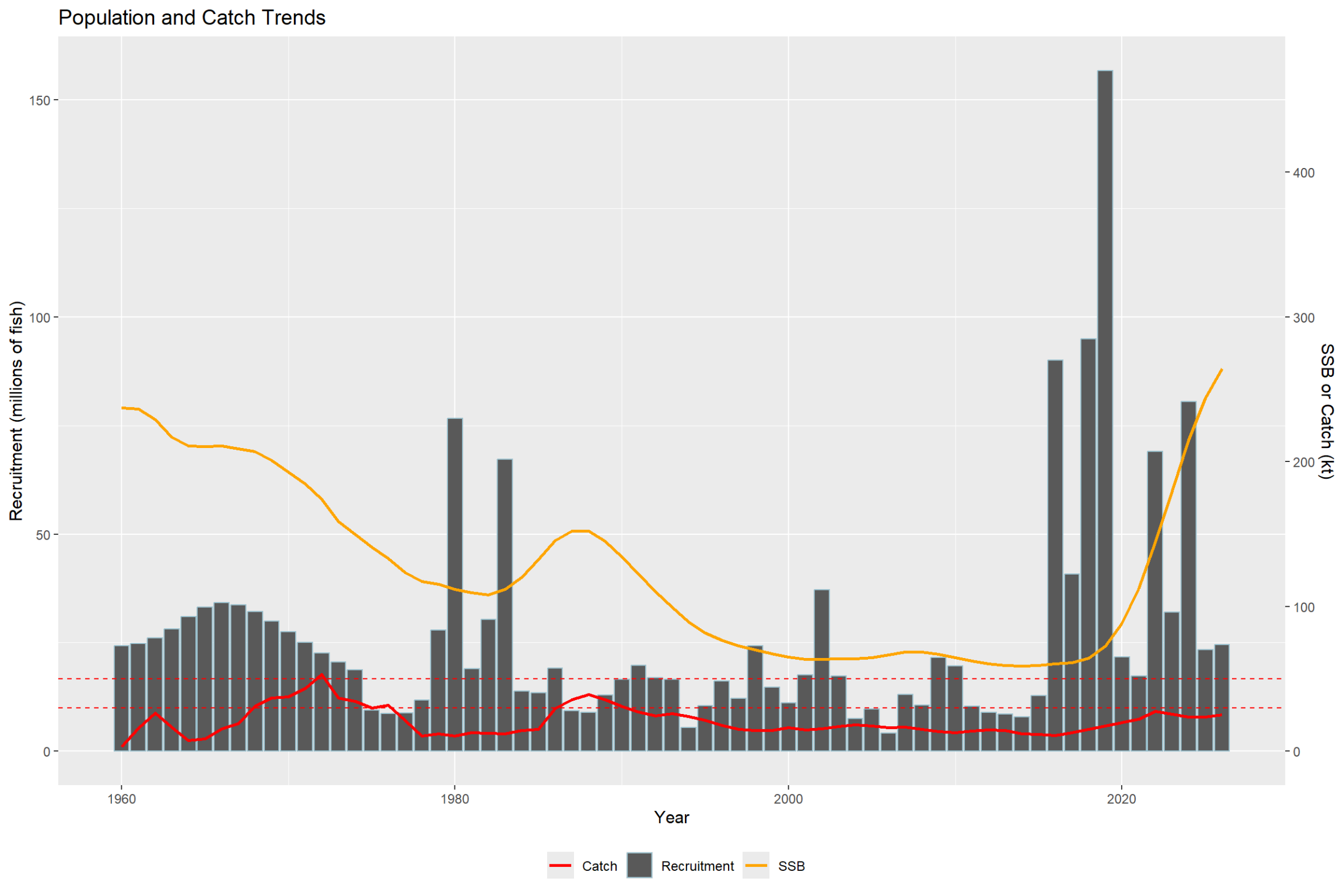

Model
26.2_FAA_M_Pr_0.115
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year

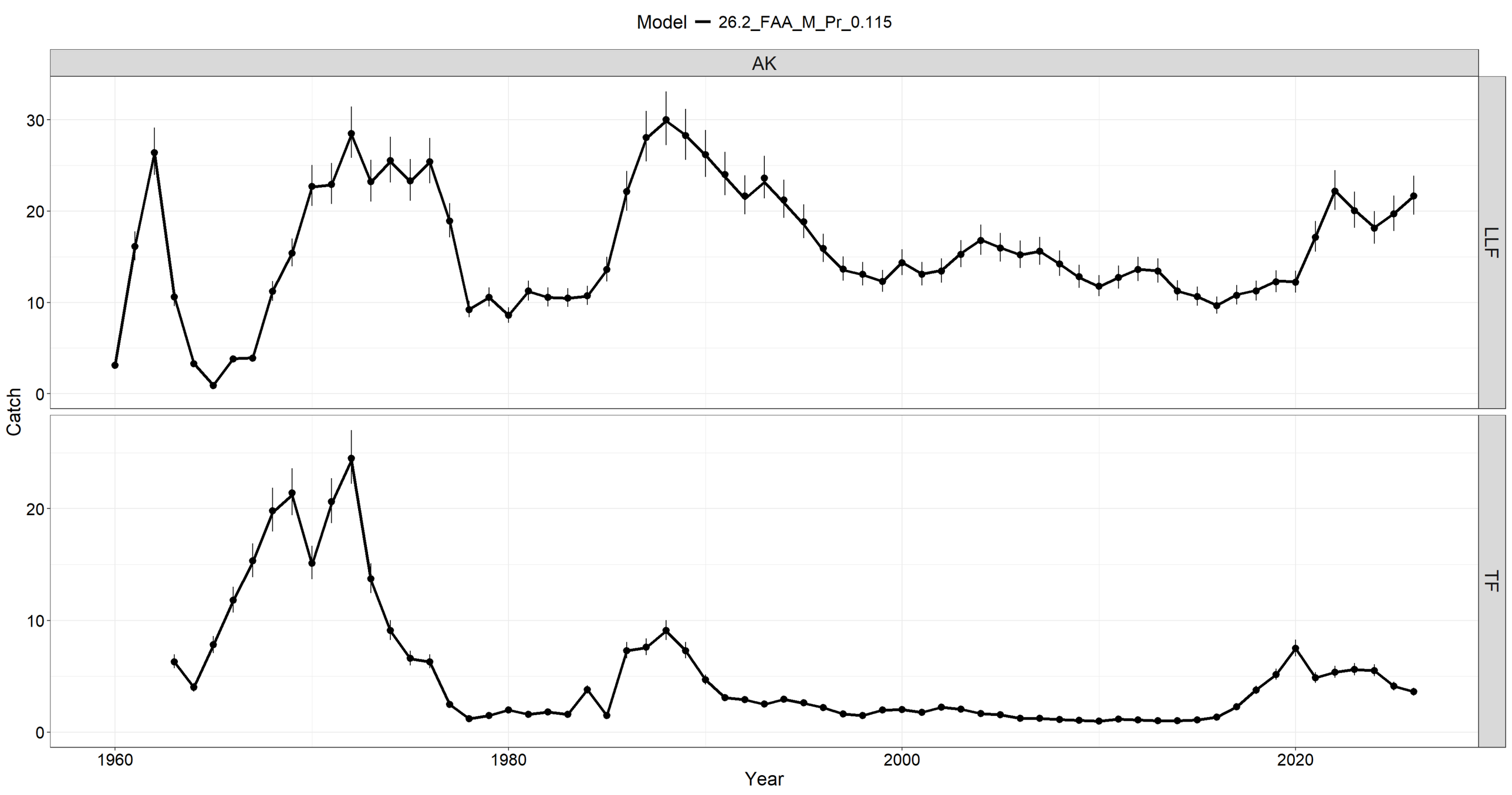

Model
26.2_FAA_M_Pr_0.115
AK
Total Biom
Biomass (kt)
1000
750
500
250
0
1960
1980
2000
2020
Year


Model
26.2_FAA_M_Pr_0.115
AK
Dynamic SSB0
Unfished SSB (kt)
0
100
200
300
400
1960
1980
2000
2020
Year


Model — 26.2_FAA_M_Pr_0.115 Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year


2019
2018
2016
2024
1980
2022
1983
2017
2002
1982
1979
2023
1998
2020
1996
1995
1994
2006
1991
1990
1984
1985
1989
1973
1972
1971
1970
1963
2025
2026
150
100
50
0
0
100
200
Age-2 Recruits (millions)
SSB (kt)


Population and Catch Trends
Recruitment (millions of fish)
SSB or Catch (kt)
150
100
50
0
400
300
200
100
0
1960
1980
2000
2020
Year
Catch
Recruitment
SSB


Model — 26.2_FAA_M_Pr_0.115
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

Model — 26.2_FAA_M_Pr_0.115

BS LLS

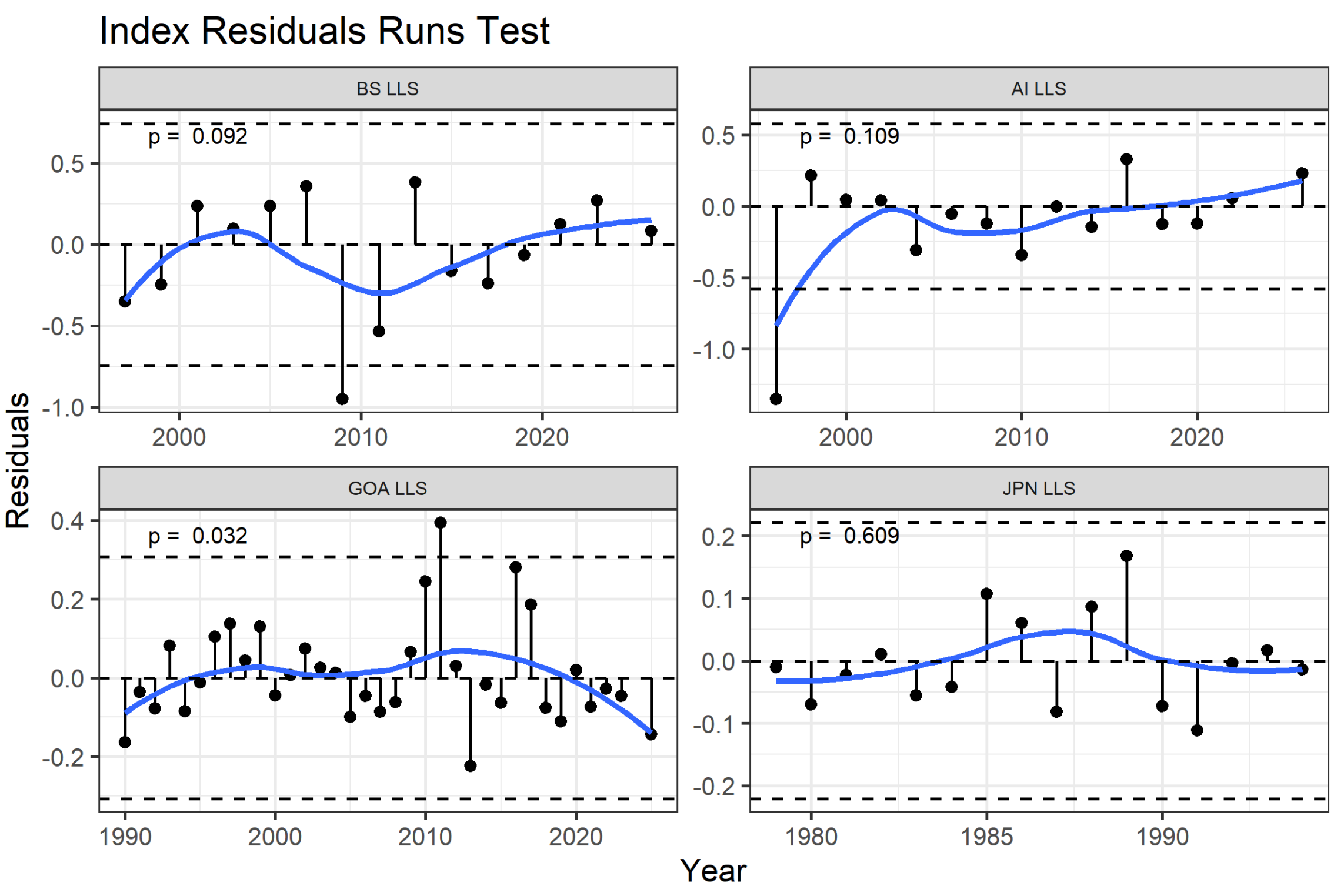
AI LLS

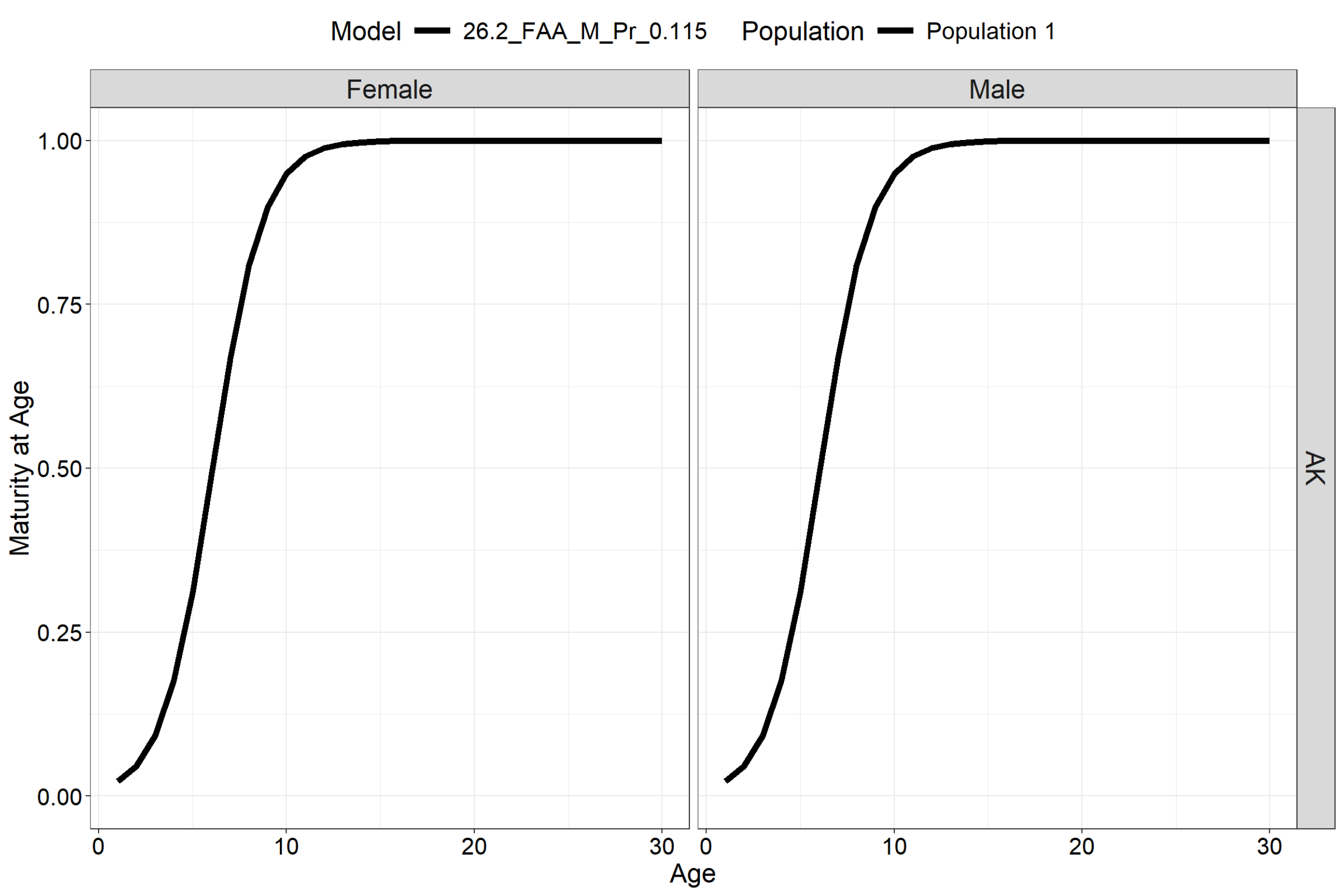
GOA LLS

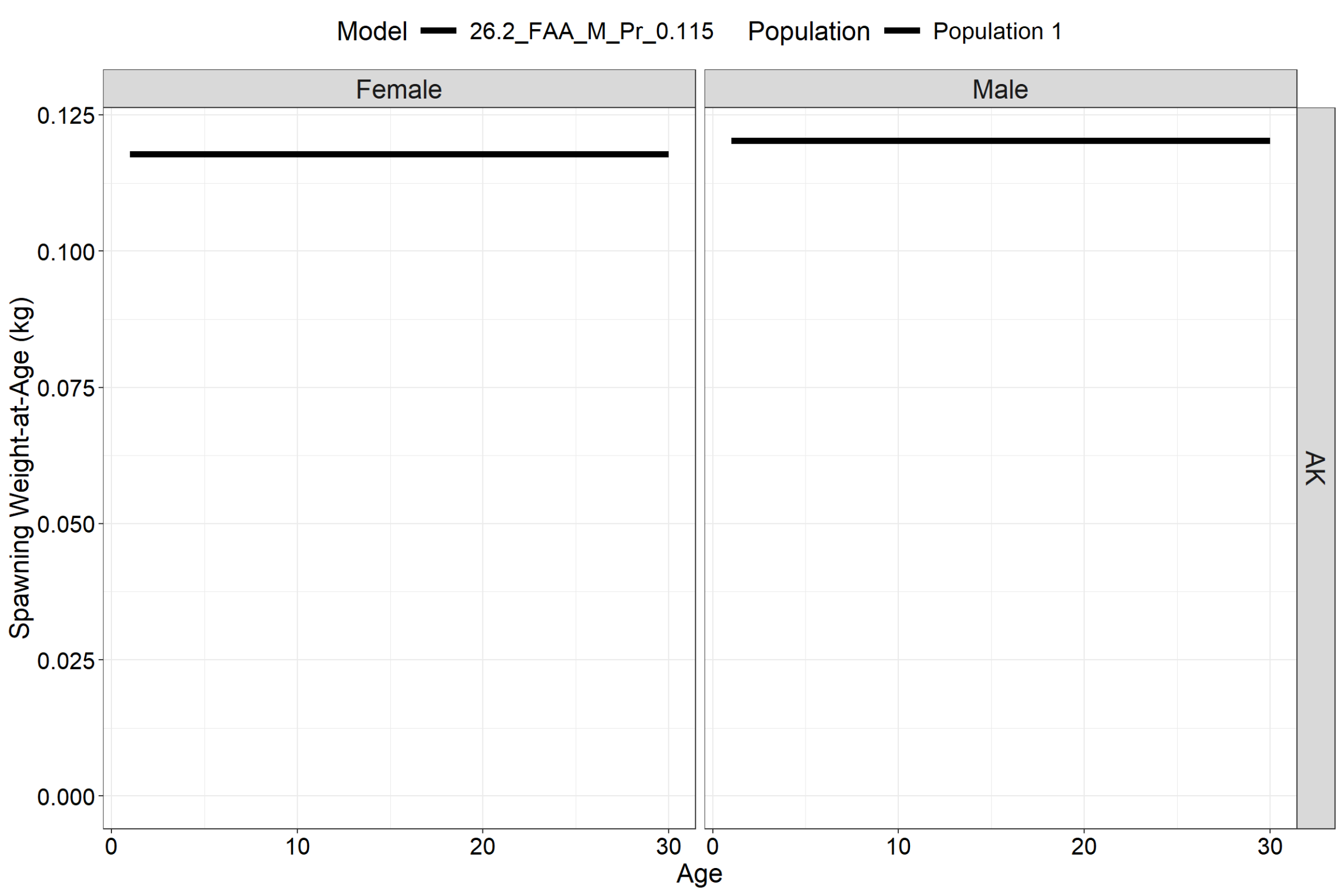
JPN LLS

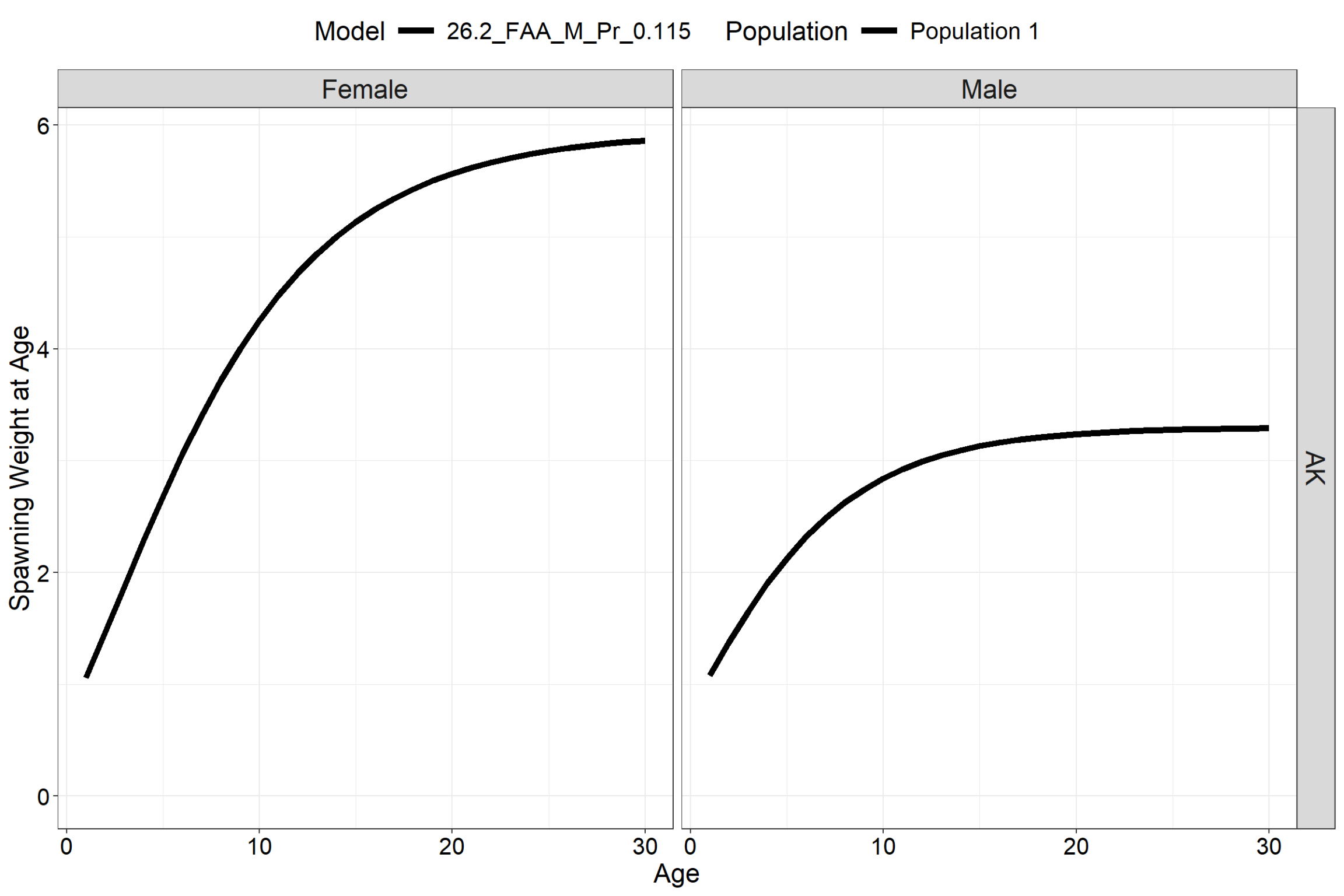
RPNs (1000s)

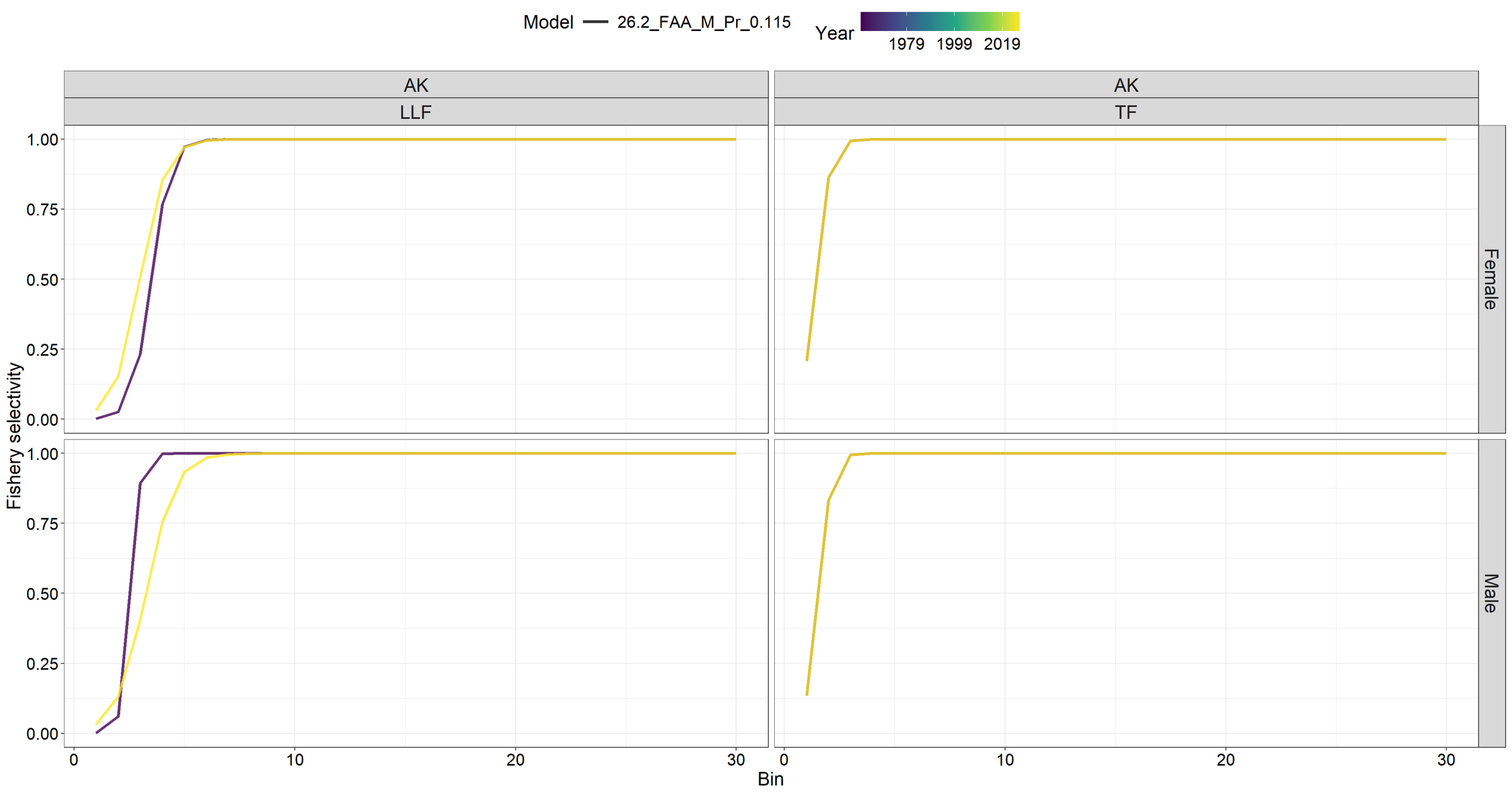
Year

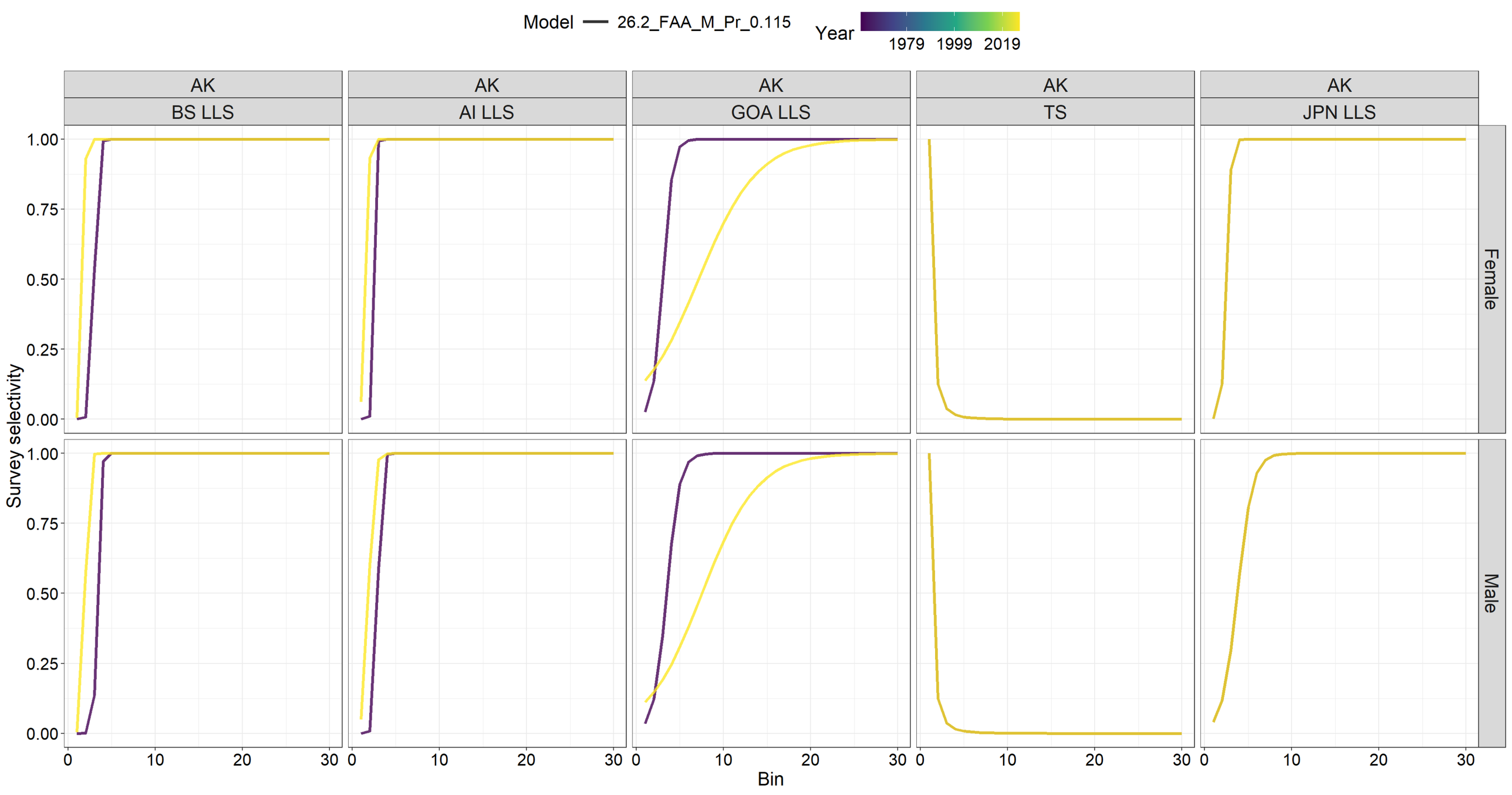

Index Residuals Runs Test
BS LLS
p = 0.092
AI LLS
p = 0.109
GOA LLS
p = 0.032
JPN LLS
p = 0.609
Residuals
Year


Model
26.2_FAA_M_Pr_0.115
Population
Population 1
Female
Male
AK
Maturity at Age
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Age


Model
26.2_FAA_M_Pr_0.115
Population
Population 1
Female
Male
AK
Spawning Weight-at-Age (kg)
0.125
0.100
0.075
0.050
0.025
0.000
0
10
20
30
Age


Model
26.2_FAA_M_Pr_0.115
Population
Population 1
Female
Male
AK
Spawning Weight at Age
Age
0
2
4
6
0
10
20
30


Model
26.2_FAA_M_Pr_0.115
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA_M_Pr_0.115
Year
1979
1999
2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin

Phase Plane

1.0
0.5
0.0

0
1
2

$F/F_{35\%}$

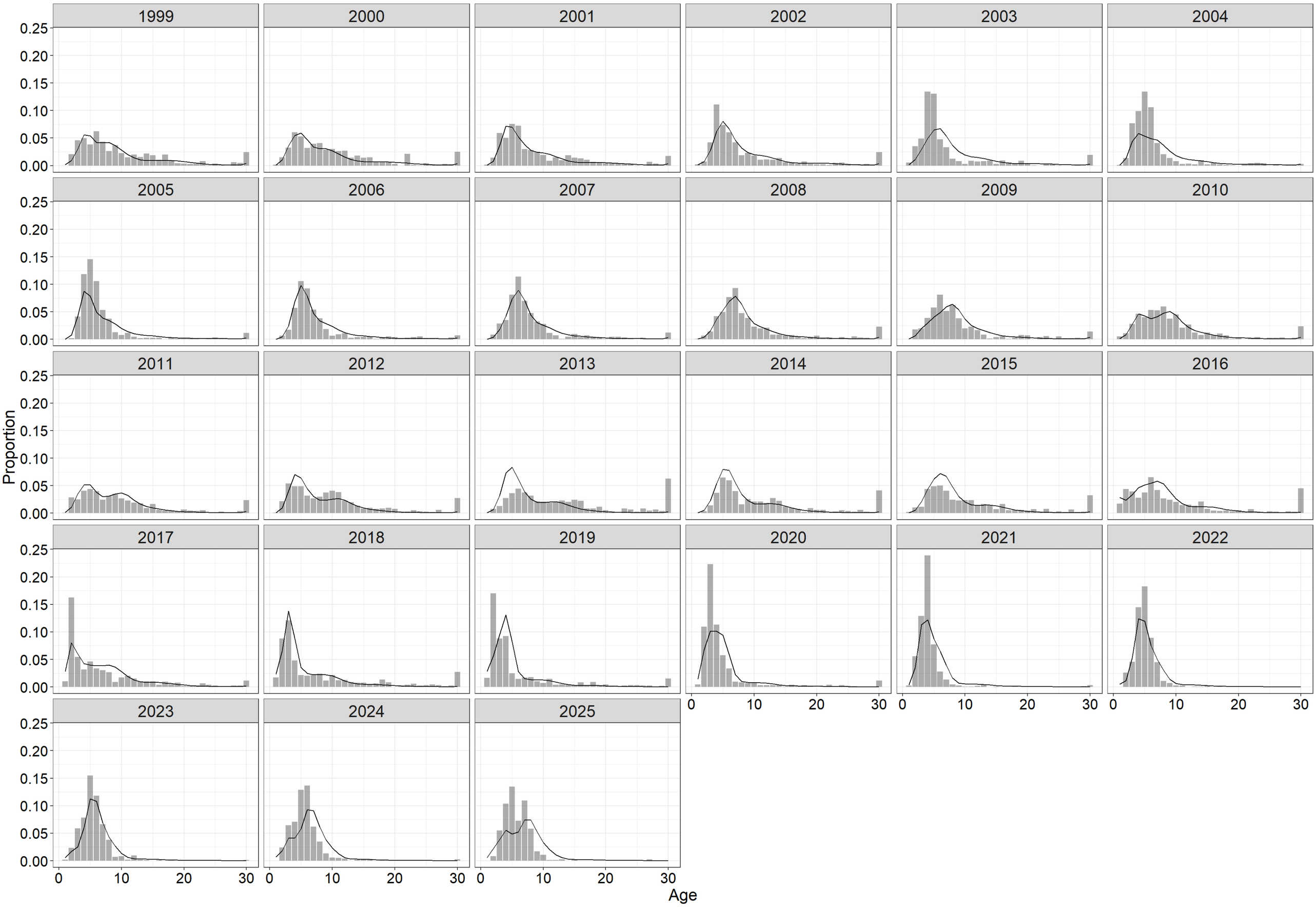
$SSB/B_{35\%}$

1995
1990
2000
2005
2010
1975
2015
1970
2020
1980
1985
2026
2025
1965
1960

Year
2020
2000
1980
1960

Control Rule
OFL (F35%)
Max ABC (F40%)

# Fixed Gear Fishery Age Comps Female AK



# Fixed Gear Fishery Age Comps Male AK

1999 2000 2001 2002 2003 2004
2005 2006 2007 2008 2009 2010
2011 2012 2013 2014 2015 2016
2017 2018 2019 2020 2021 2022
2023 2024 2025

Proportion: 0.00, 0.05, 0.10, 0.15

Age: 0, 10, 20, 30

**Fixed Gear Fishery Age Comps OSA AK**

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=1.00
(0.97-1.03)

Sample quantiles

Theoretical quantiles

Female

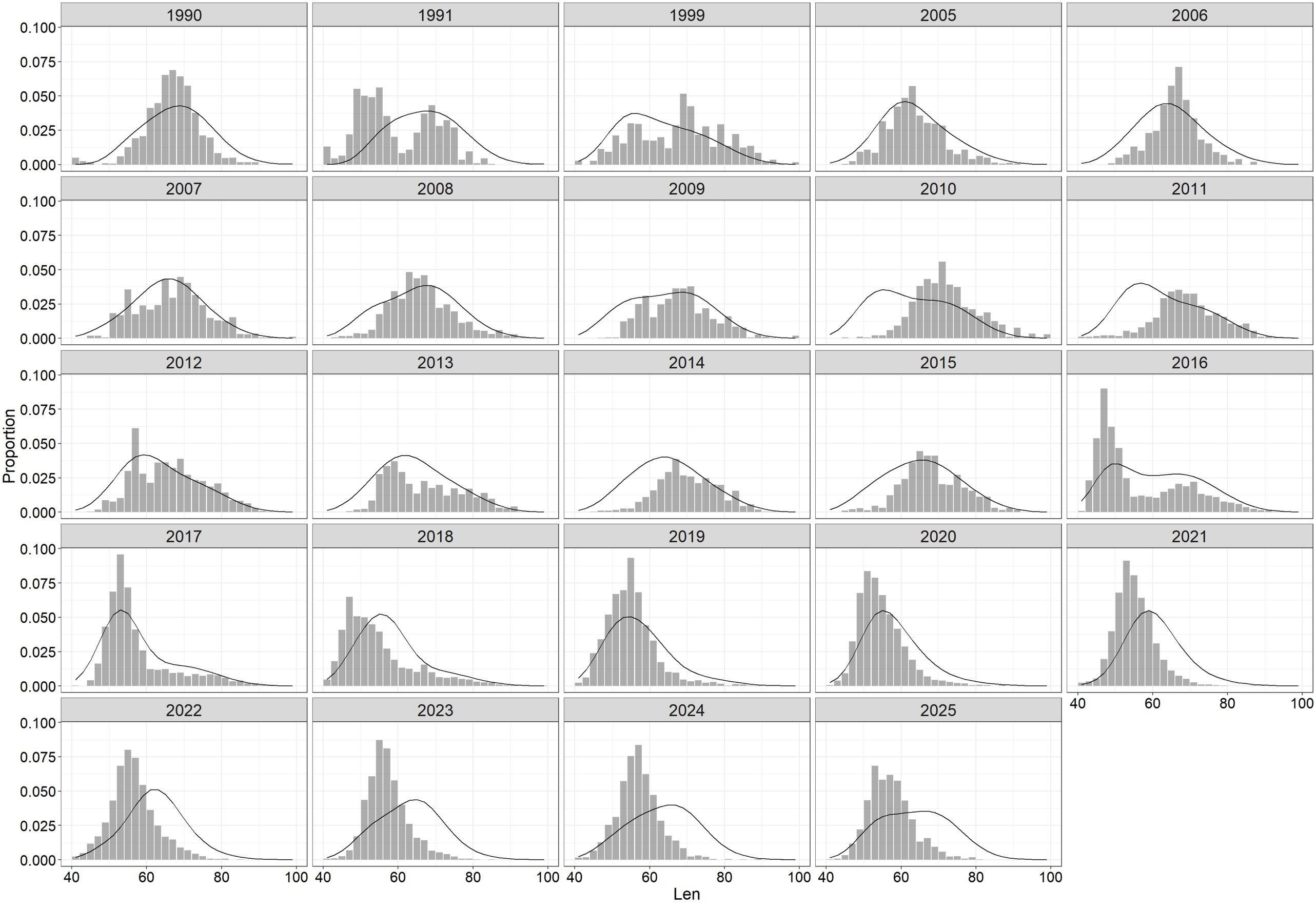
Male

AK

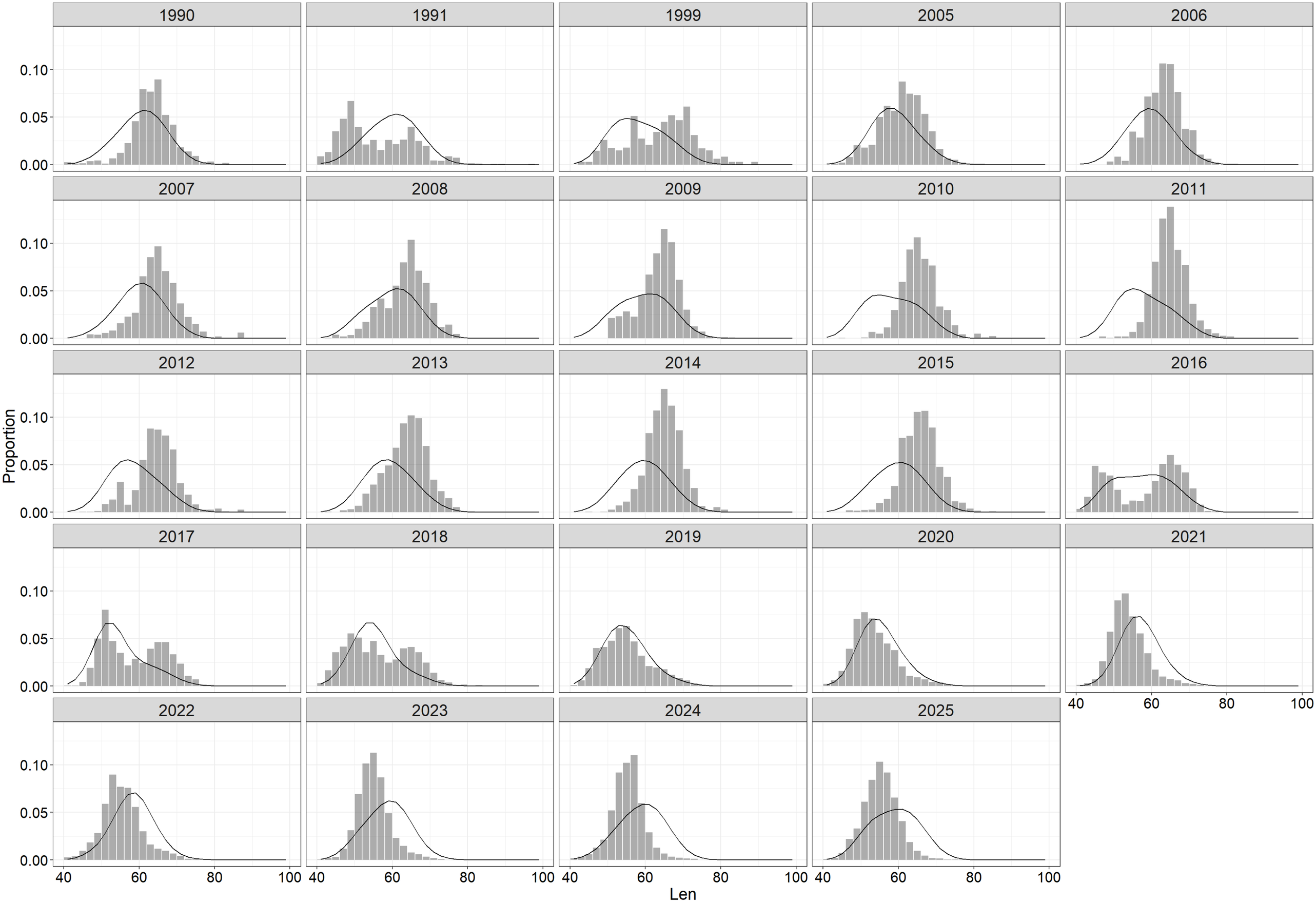
Proportions

Age

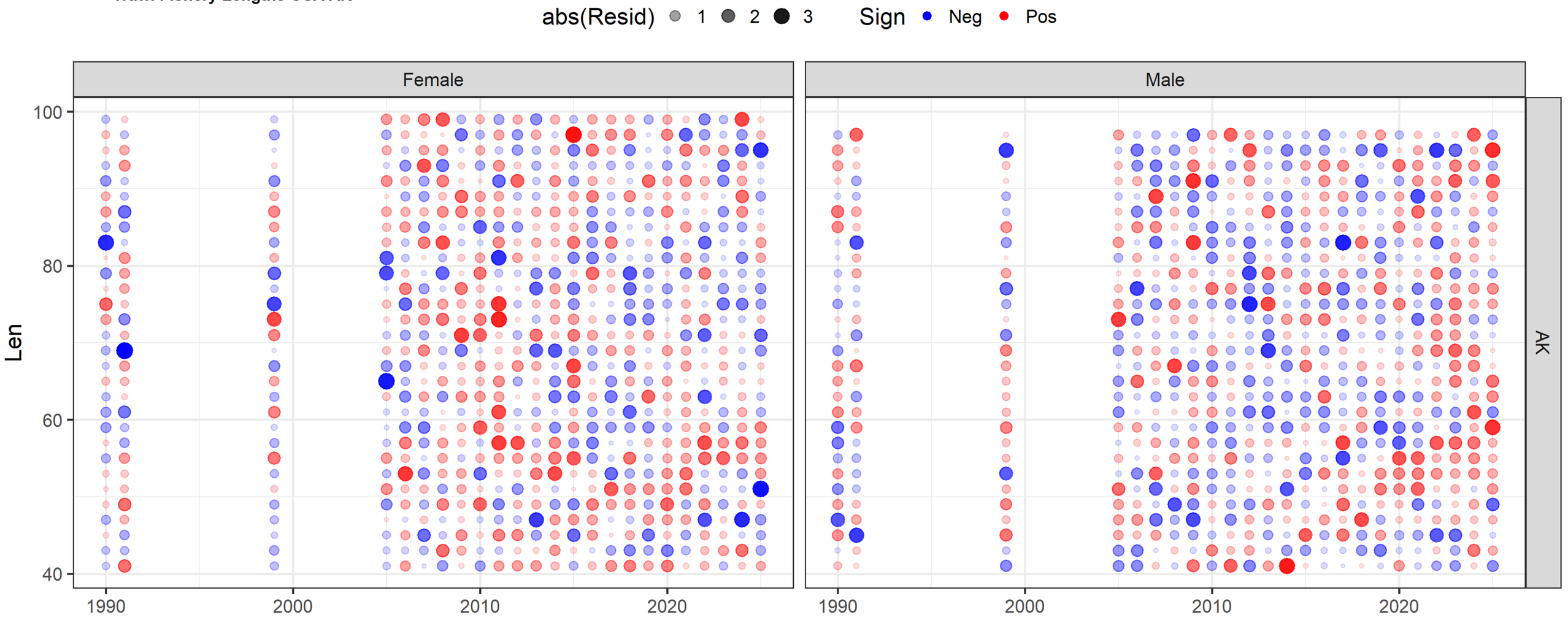
# Trawl Fishery Lengths Female AK

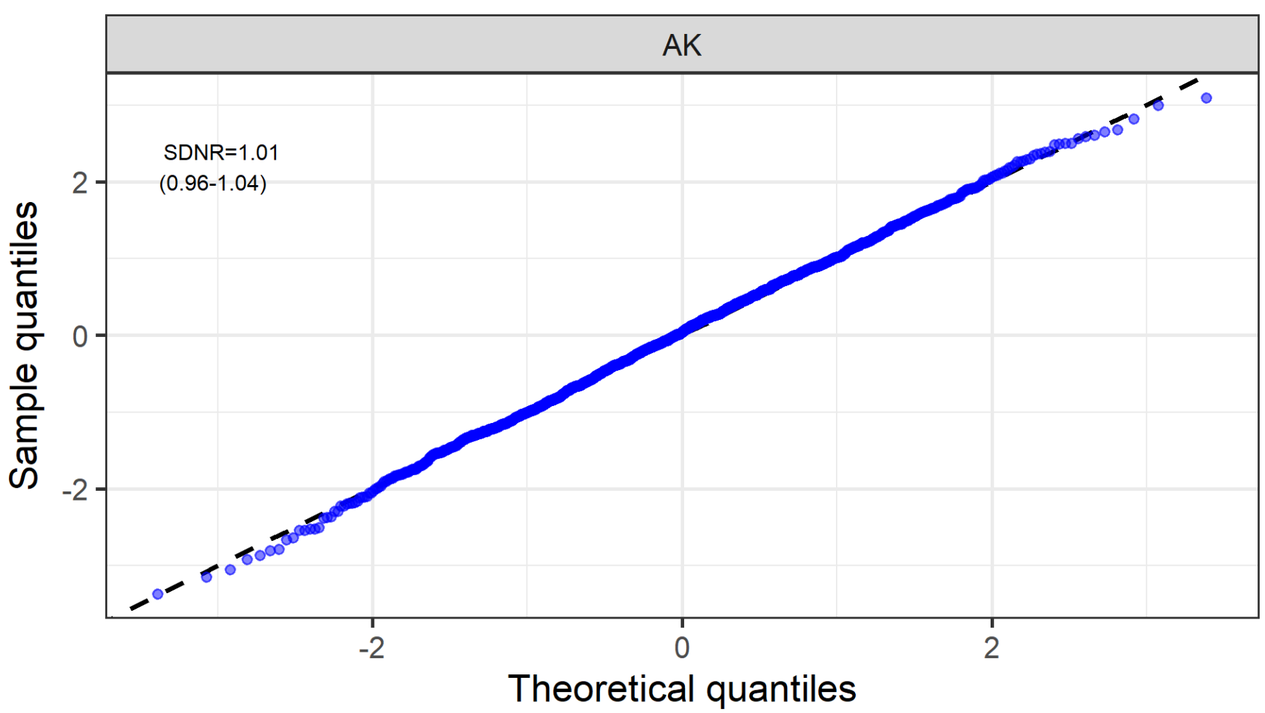


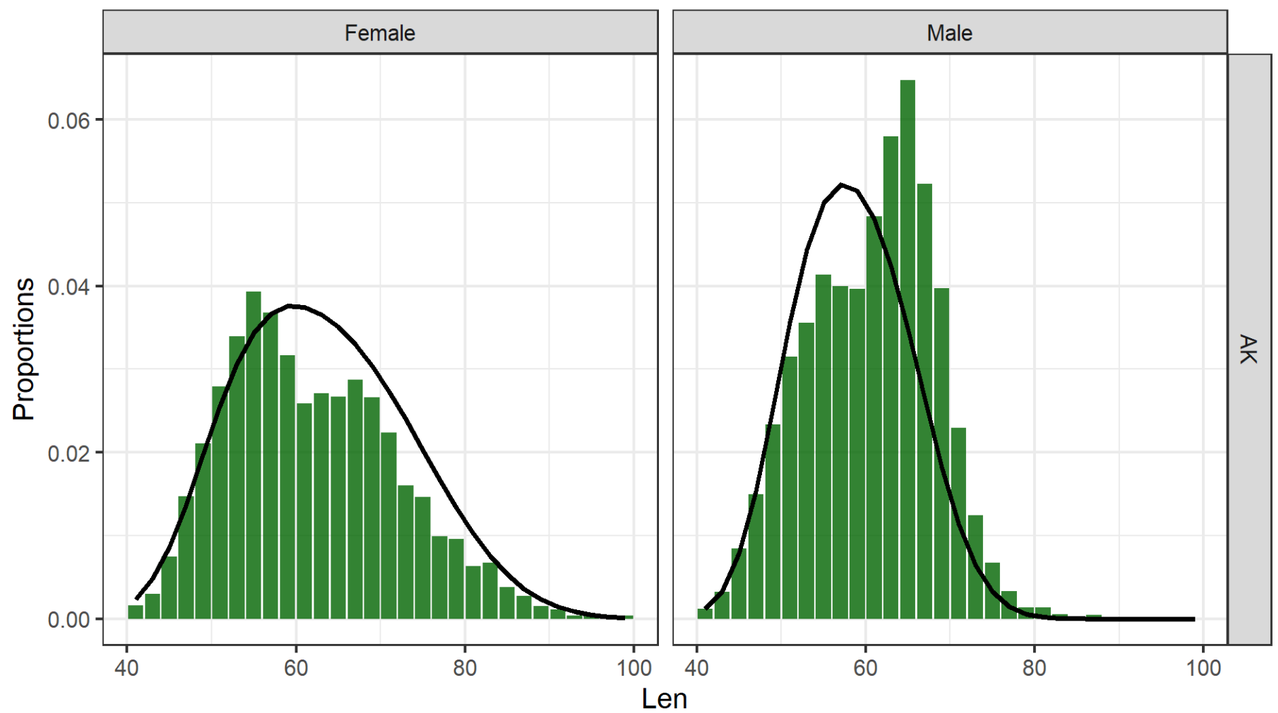
# Trawl Fishery Lengths Male AK



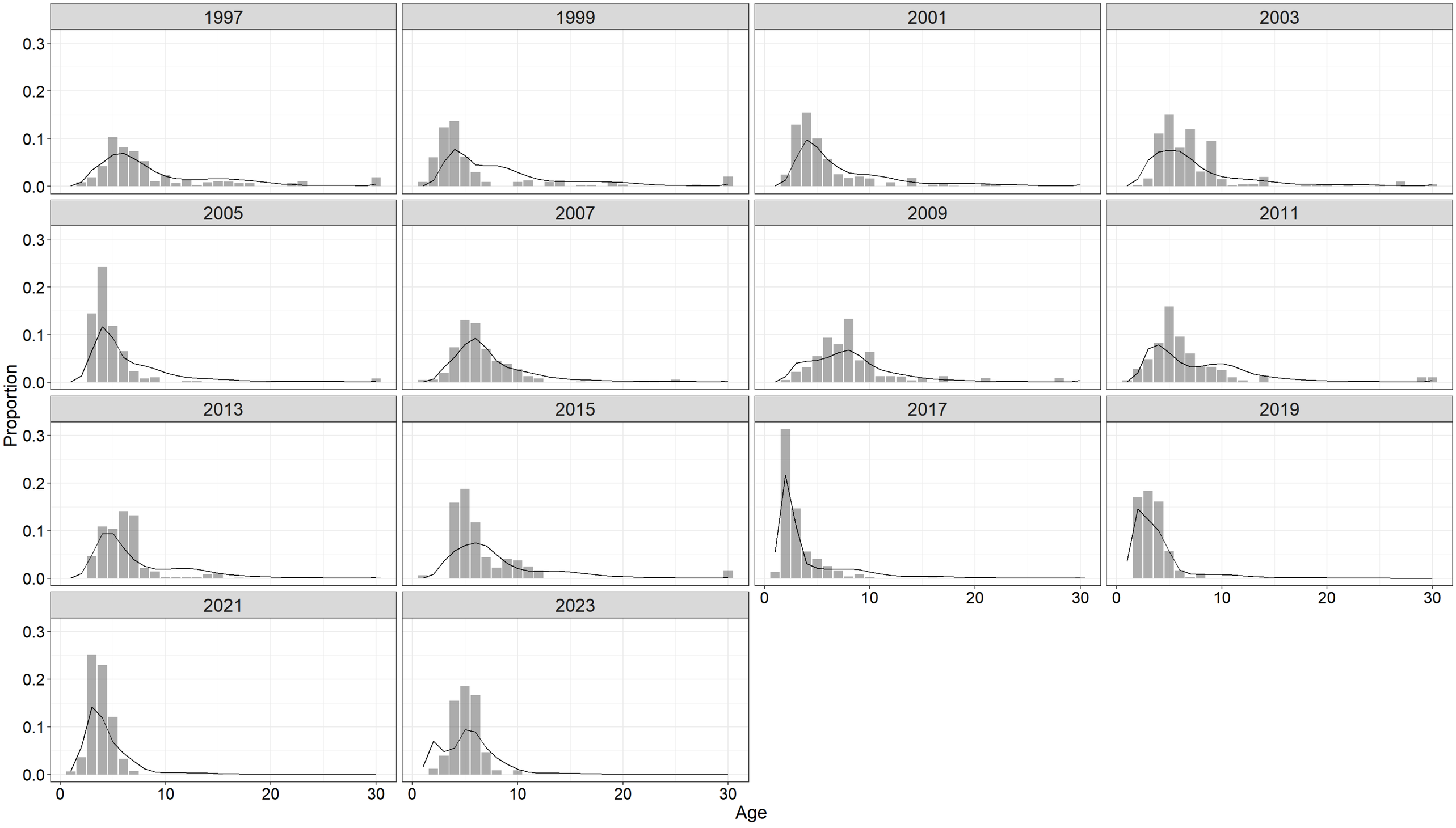
**Trawl Fishery Lengths OSA AK**



# BS LL Survey Age Comps Female AK

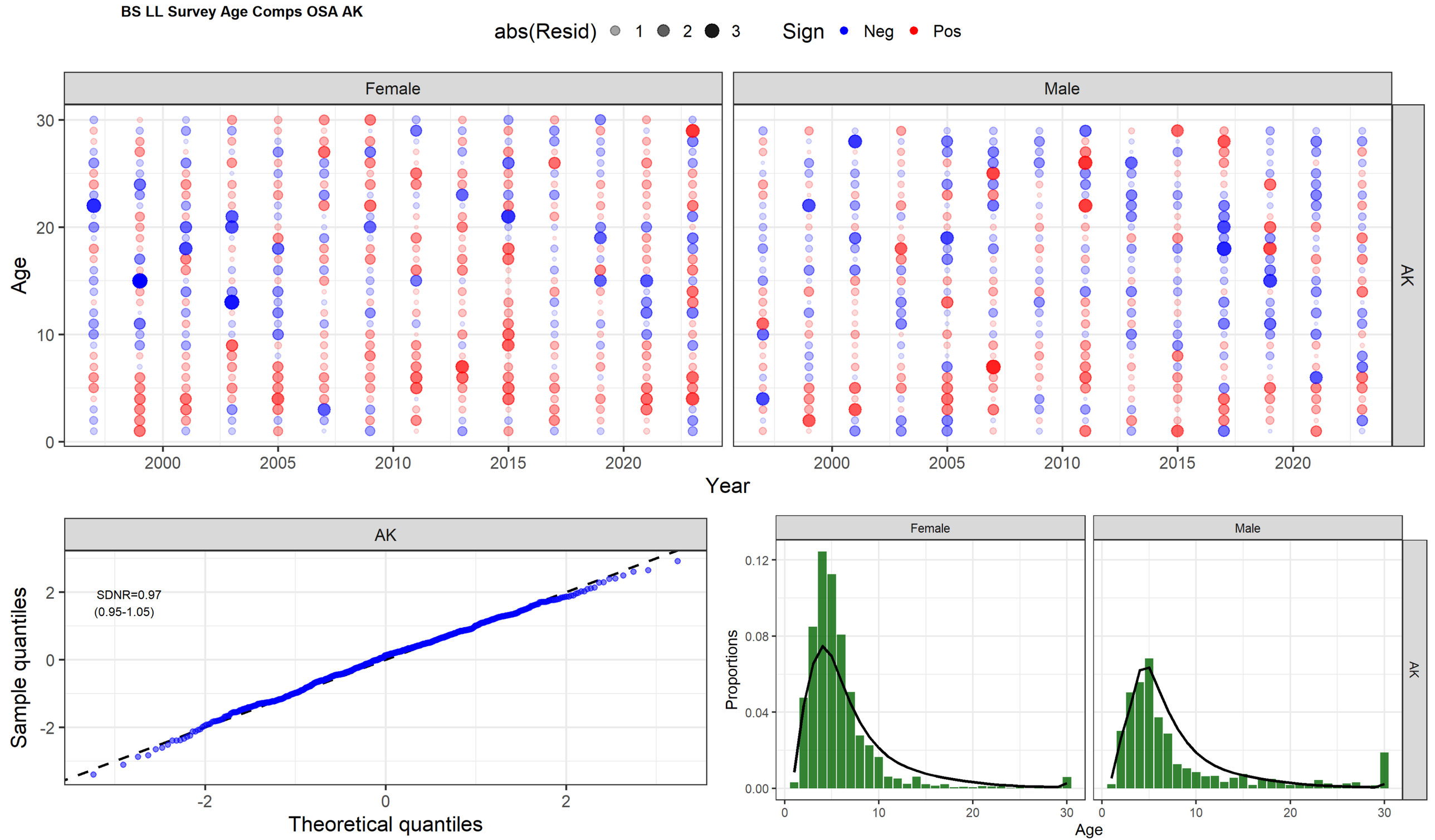


# BS LL Survey Age Comps Male AK

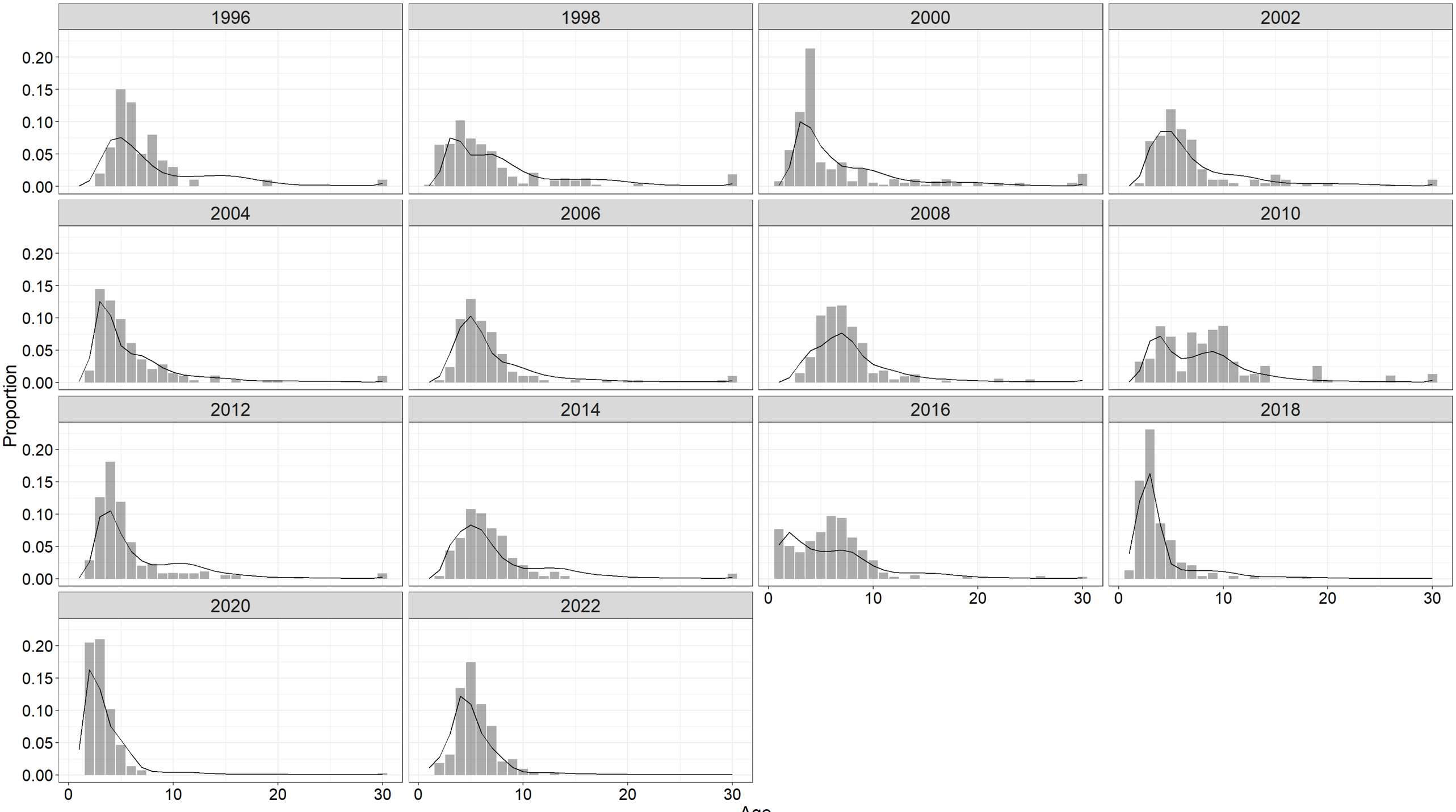
1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

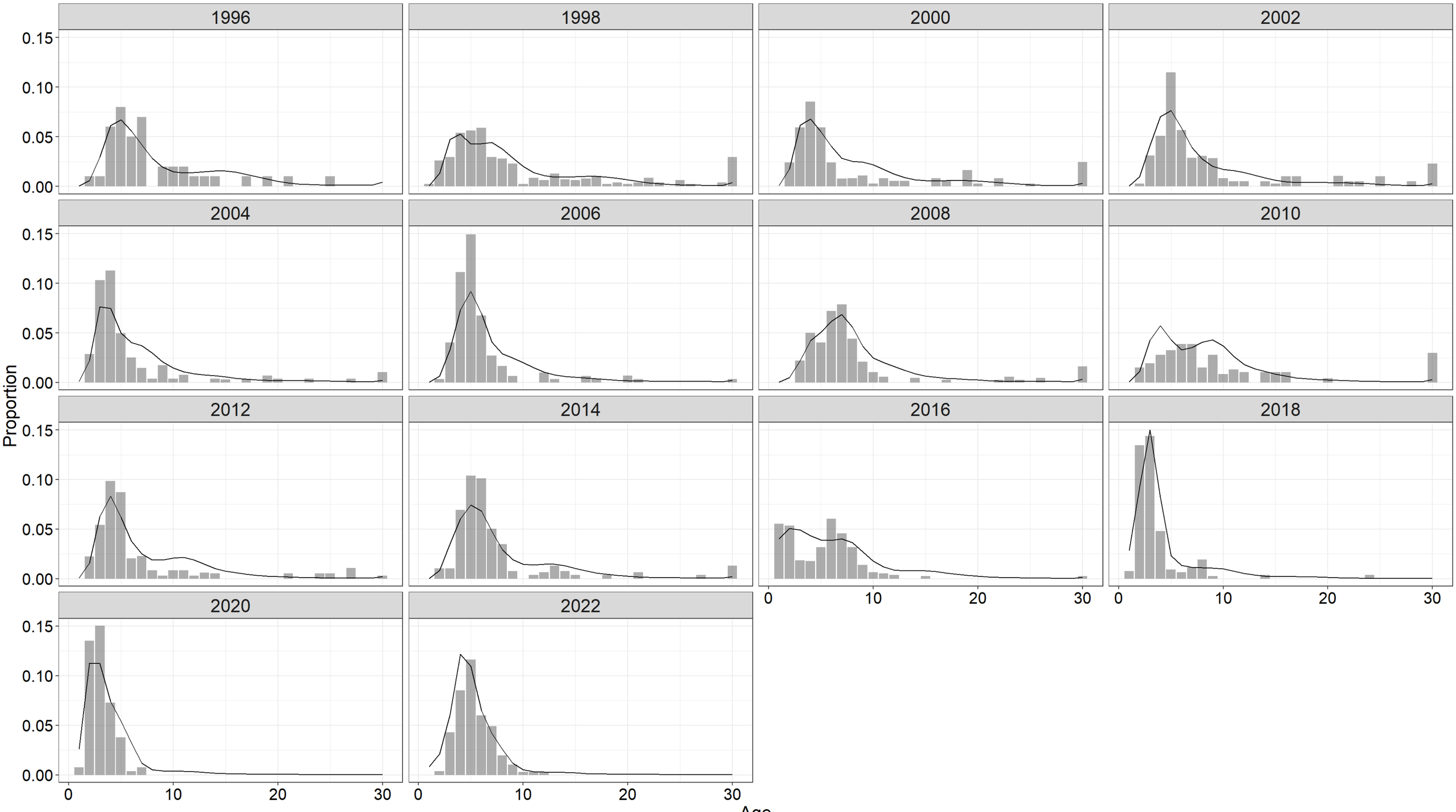
**BS LL Survey Age Comps OSA AK**



# AI LL Survey Age Comps Female AK



# AI LL Survey Age Comps Male AK



AI LL Survey Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=0.96
(0.95-1.05)

Sample quantiles

Theoretical quantiles

Female

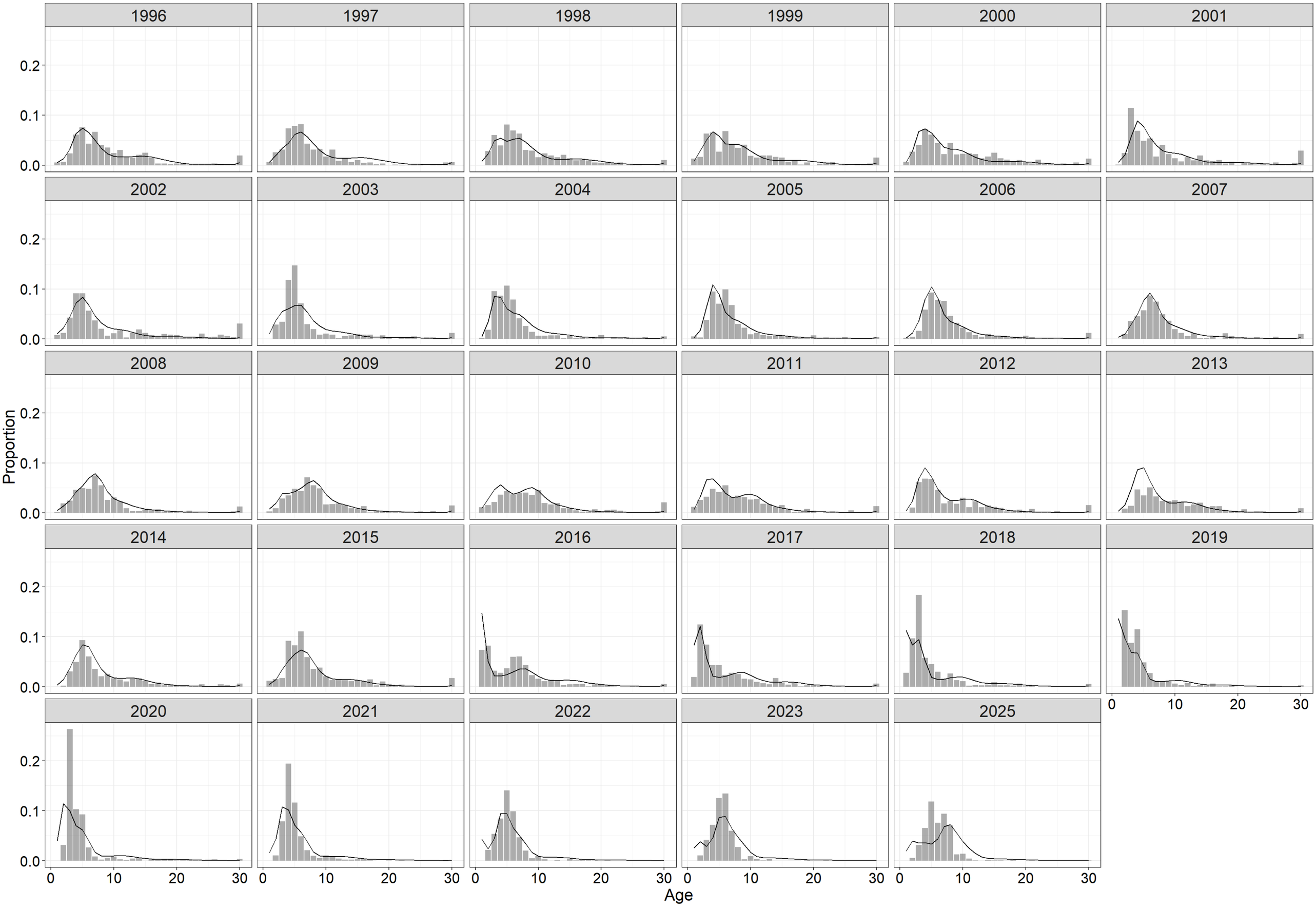
Male

AK

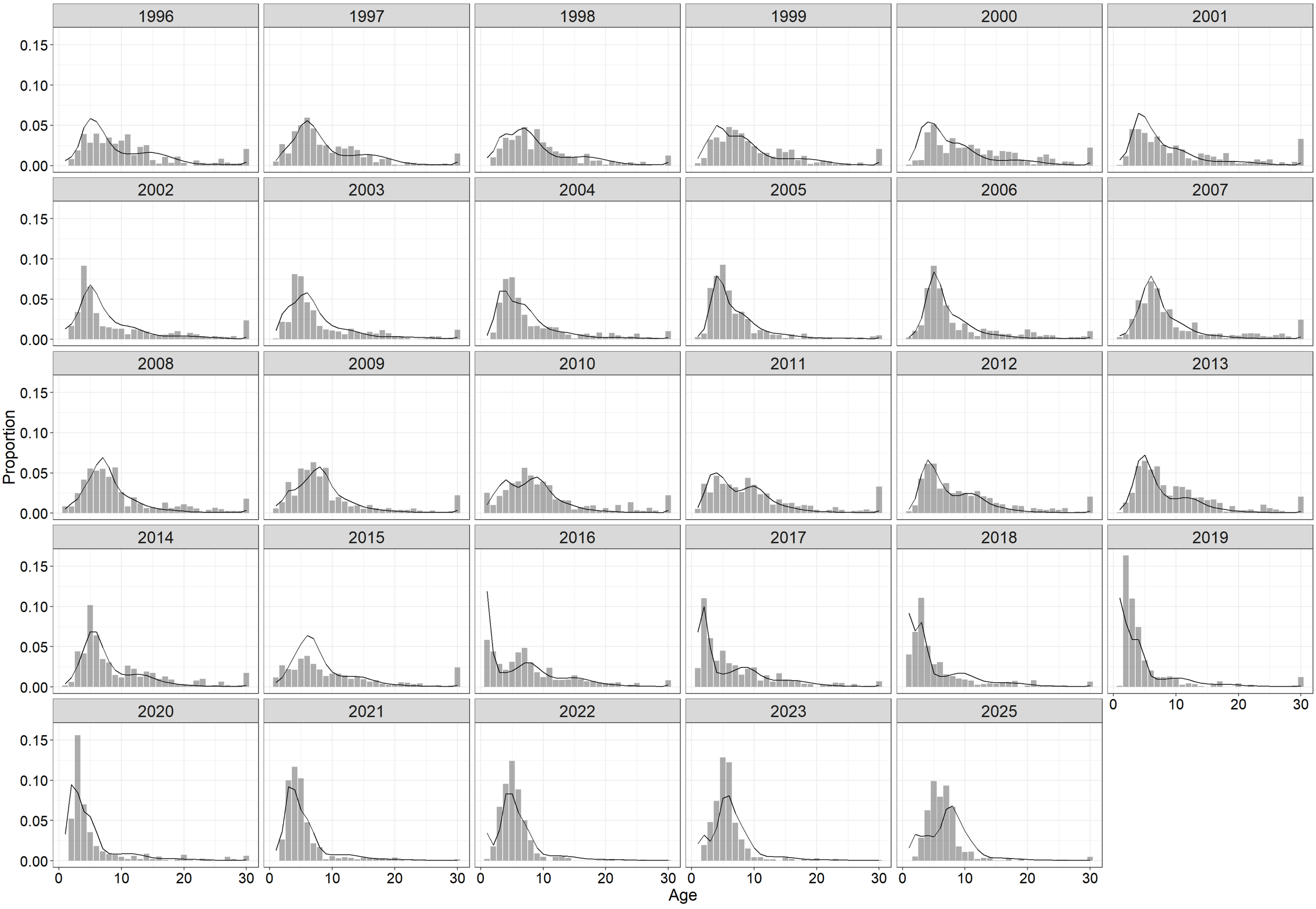
Proportions

Age

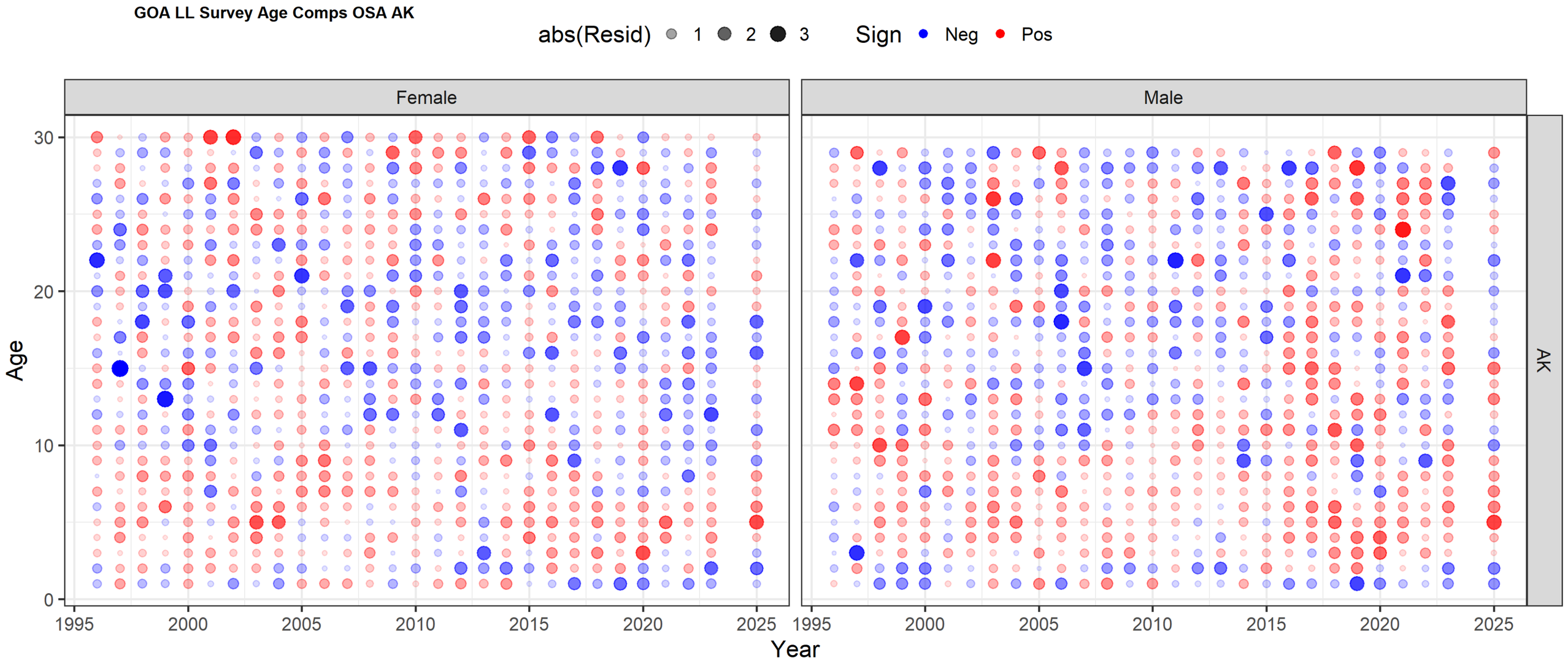
# GOA LL Survey Age Comps Female AK

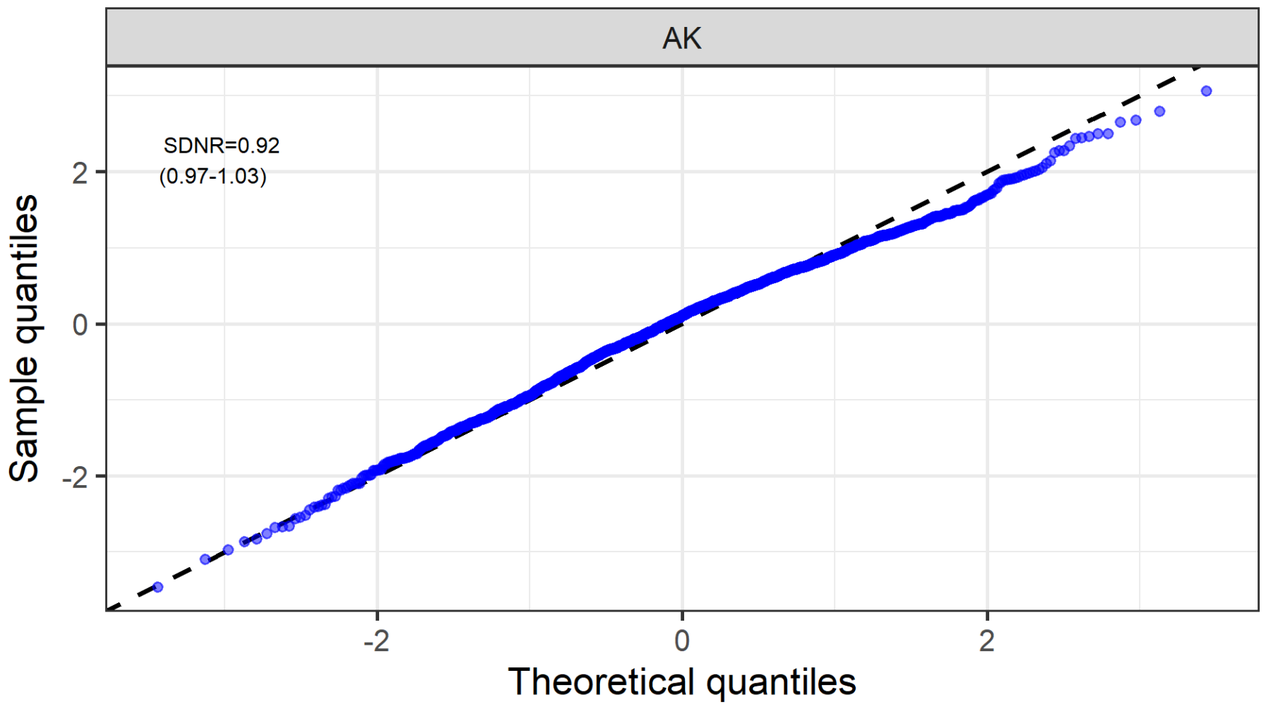


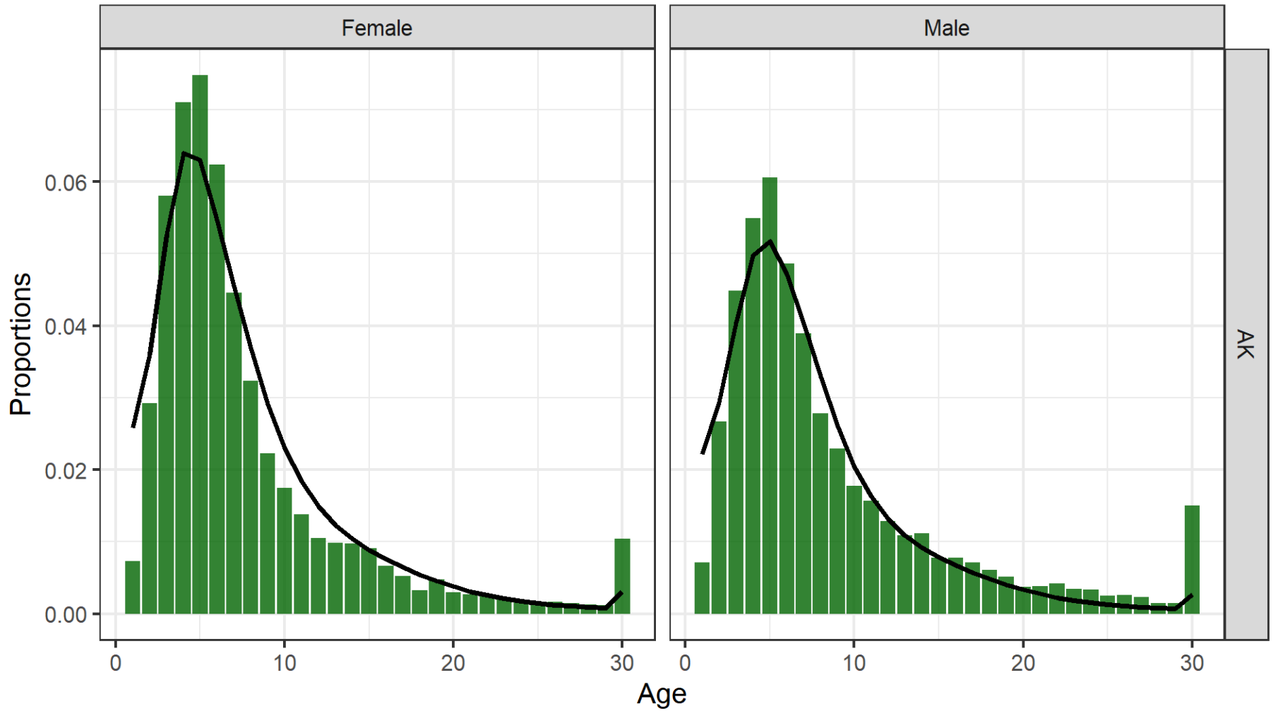
# GOA LL Survey Age Comps Male AK

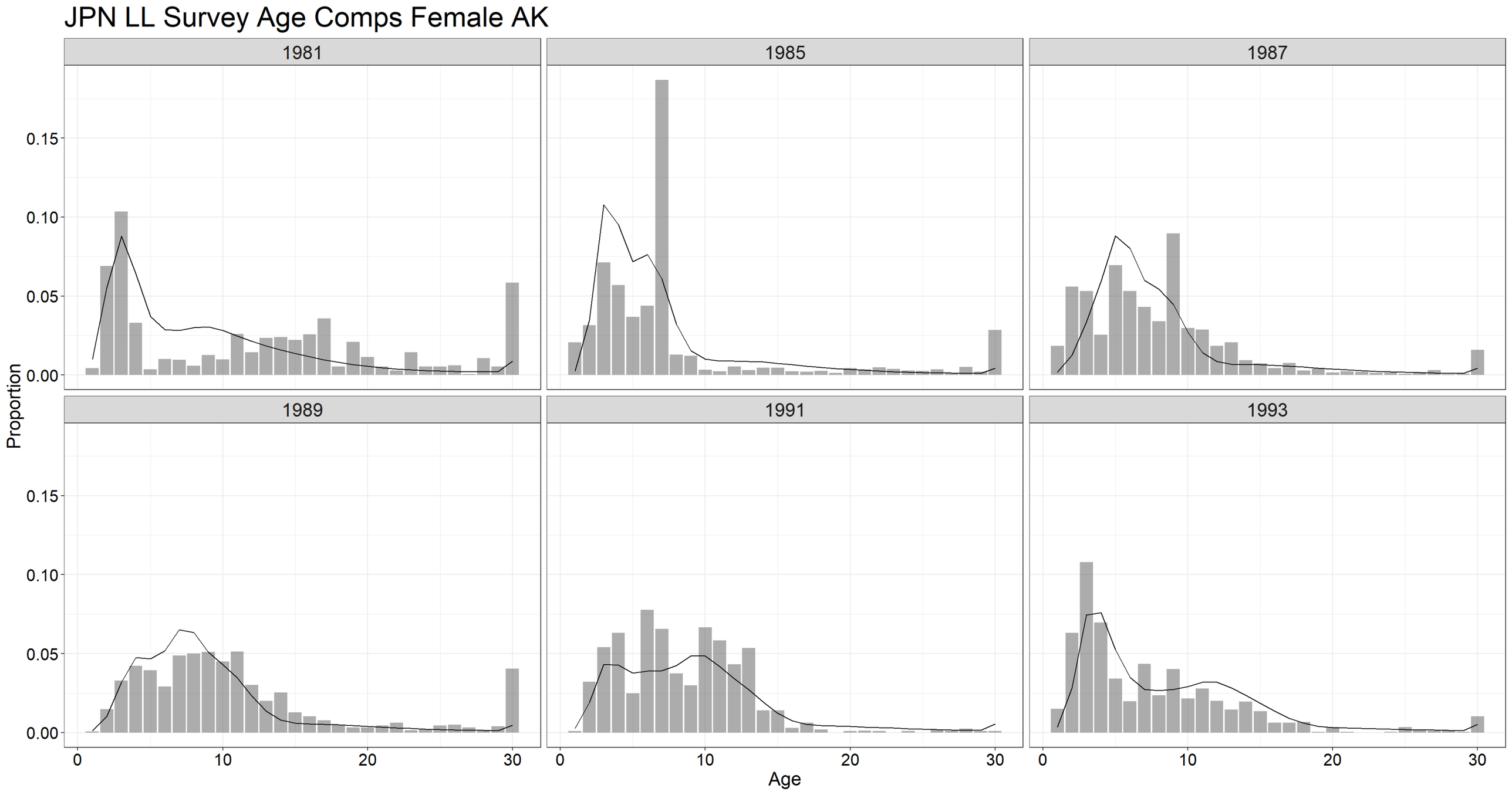


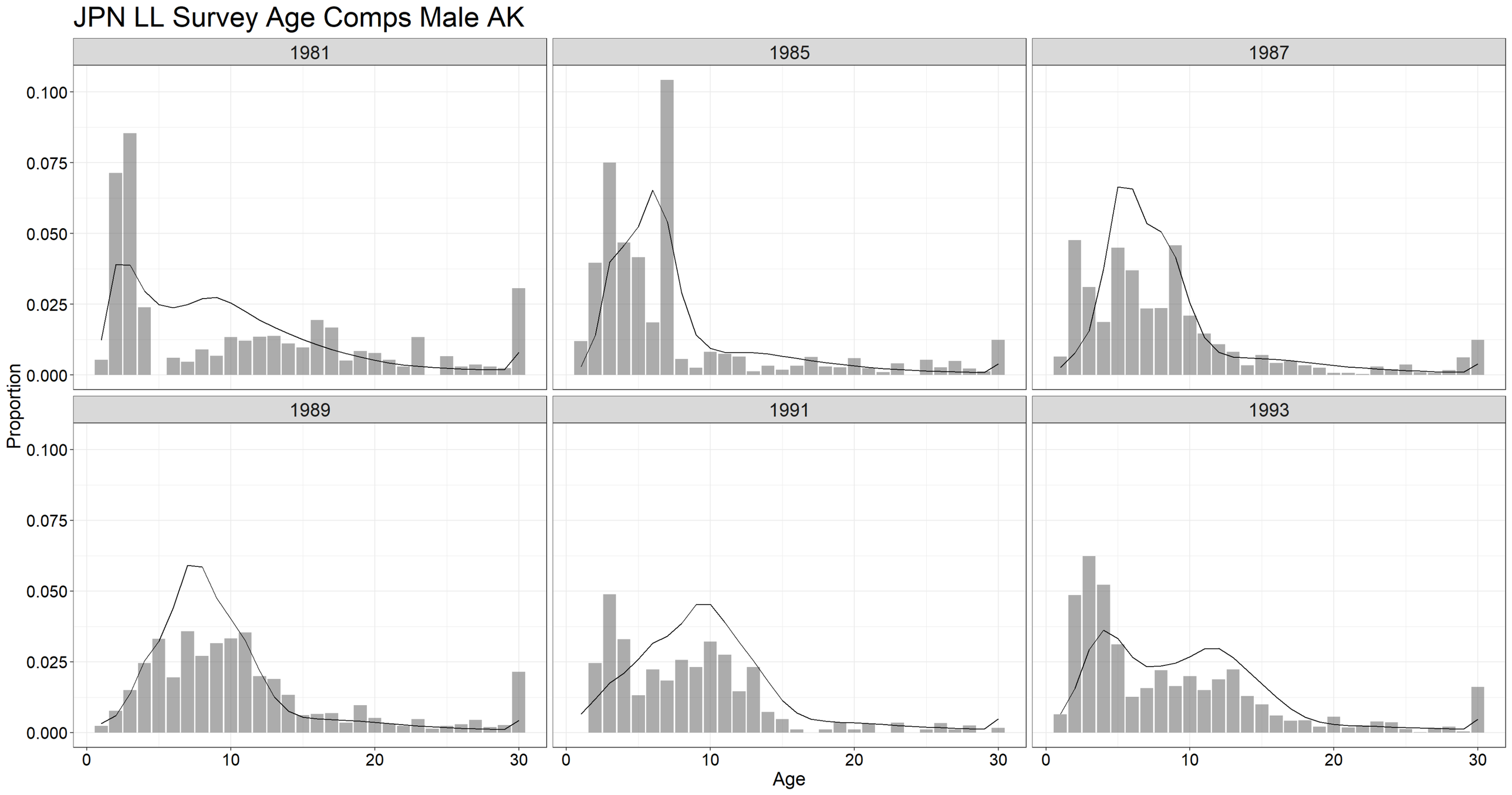
**GOA LL Survey Age Comps OSA AK**

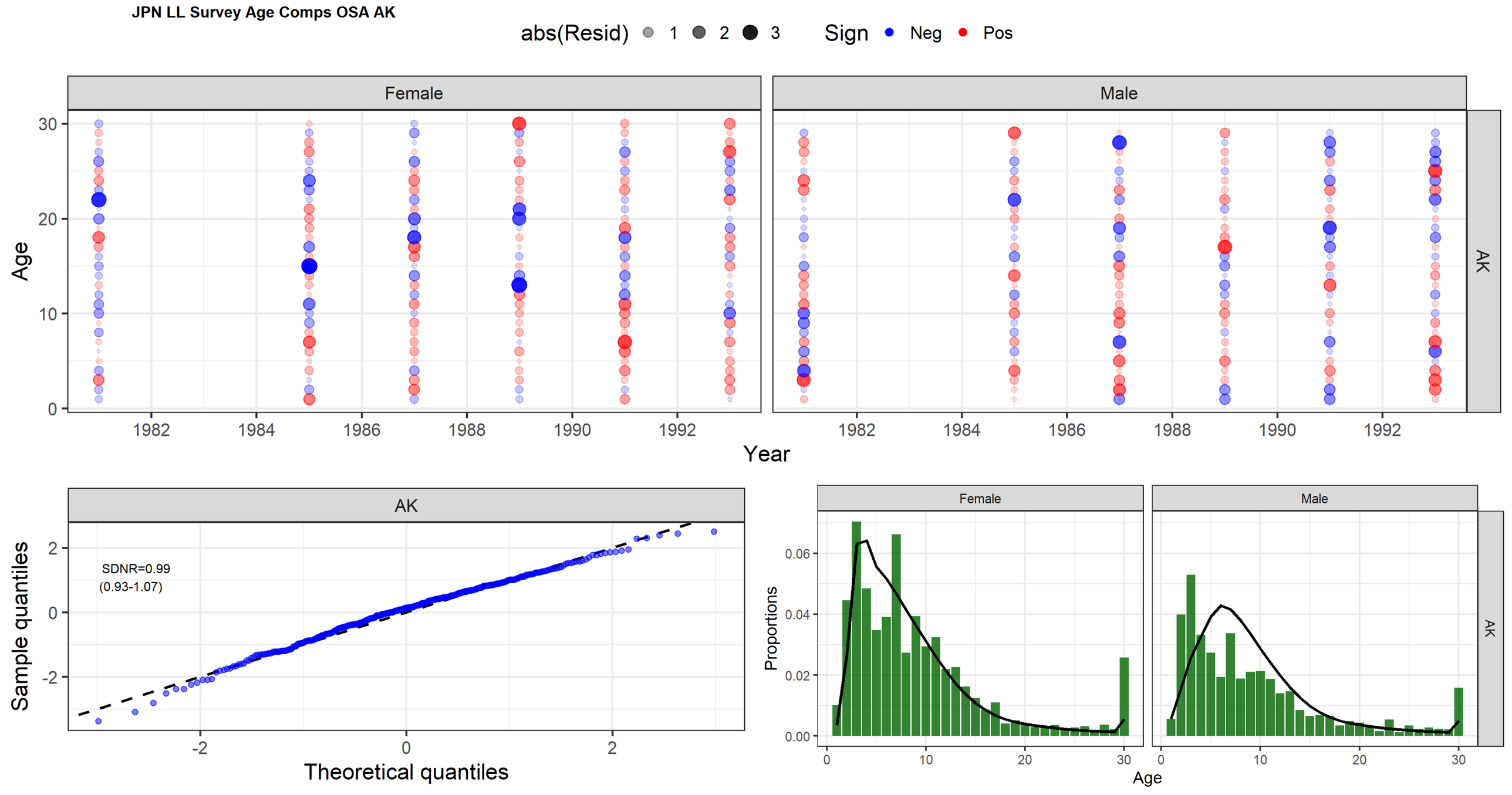



JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
Proportion
0.100
0.075
0.050
0.025
0.000
0
10
20
30
Age


JPN LL Survey Age Comps OSA AK
abs(Resid) 1 2 3
Sign Neg Pos
Female
Male
AK
Age
Year
SDNR=0.99
(0.93-1.07)
Sample quantiles
Theoretical quantiles
Proportions
Age

# JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

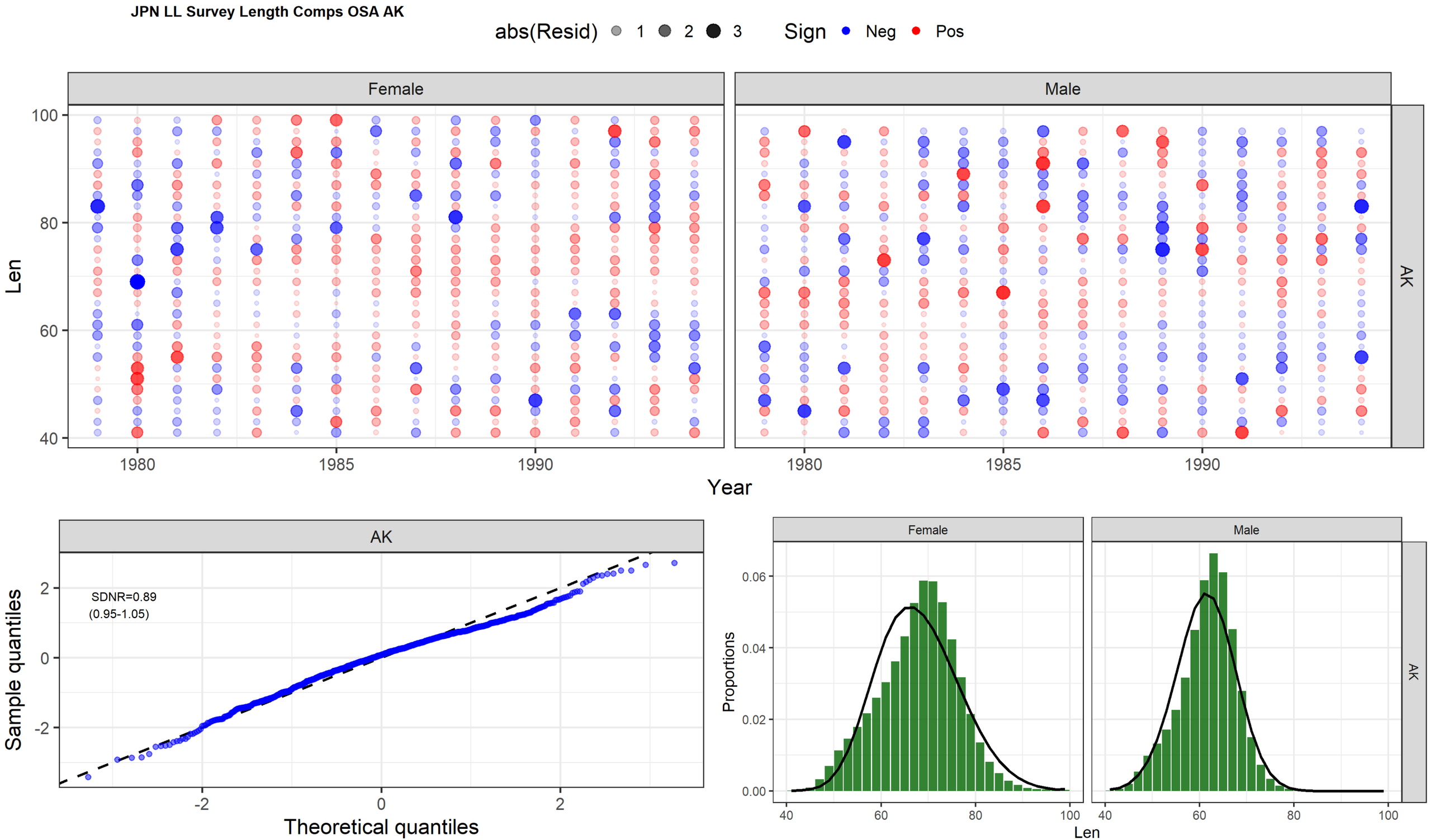
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

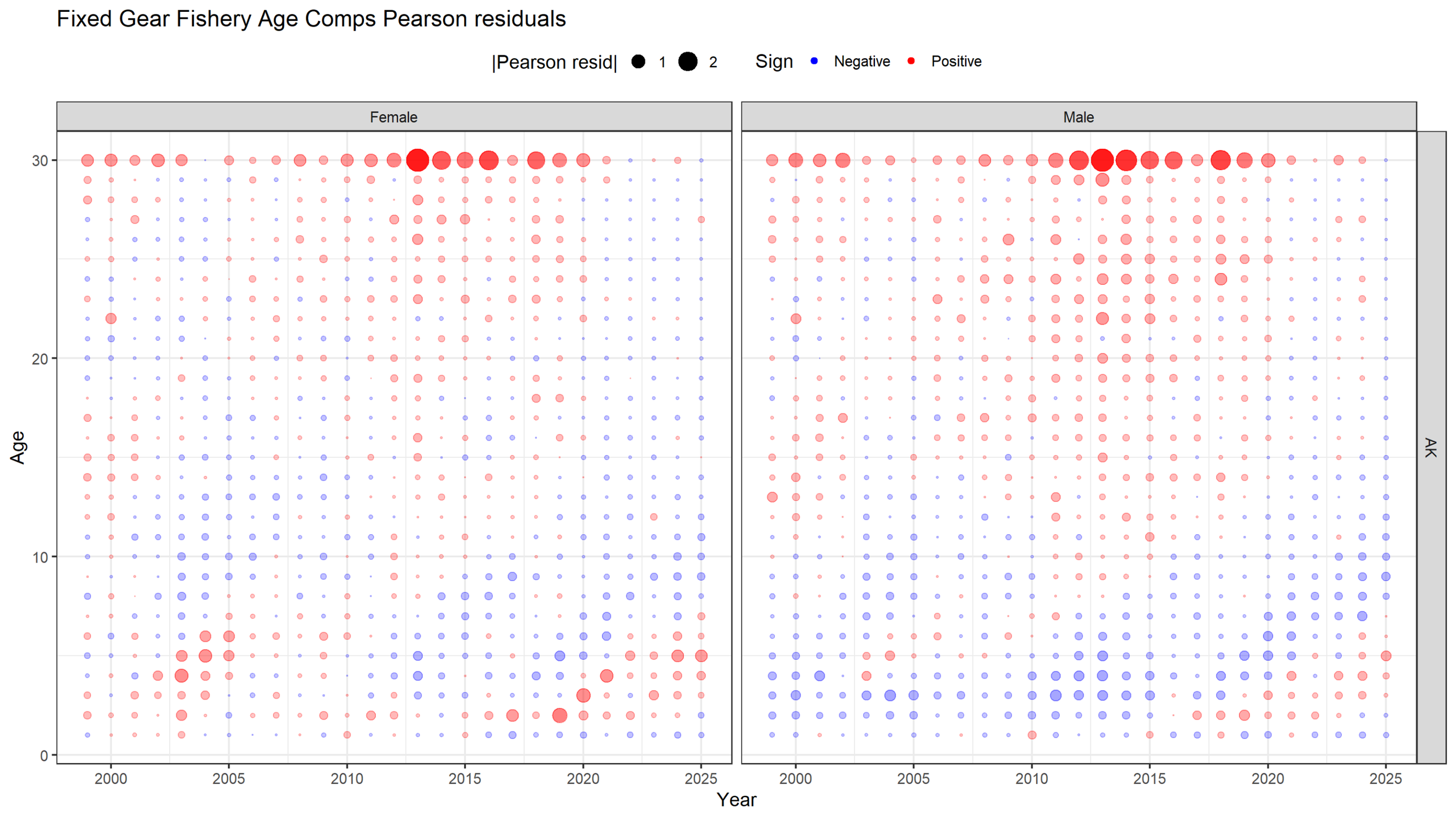
# JPN LL Survey Length Comps Male AK

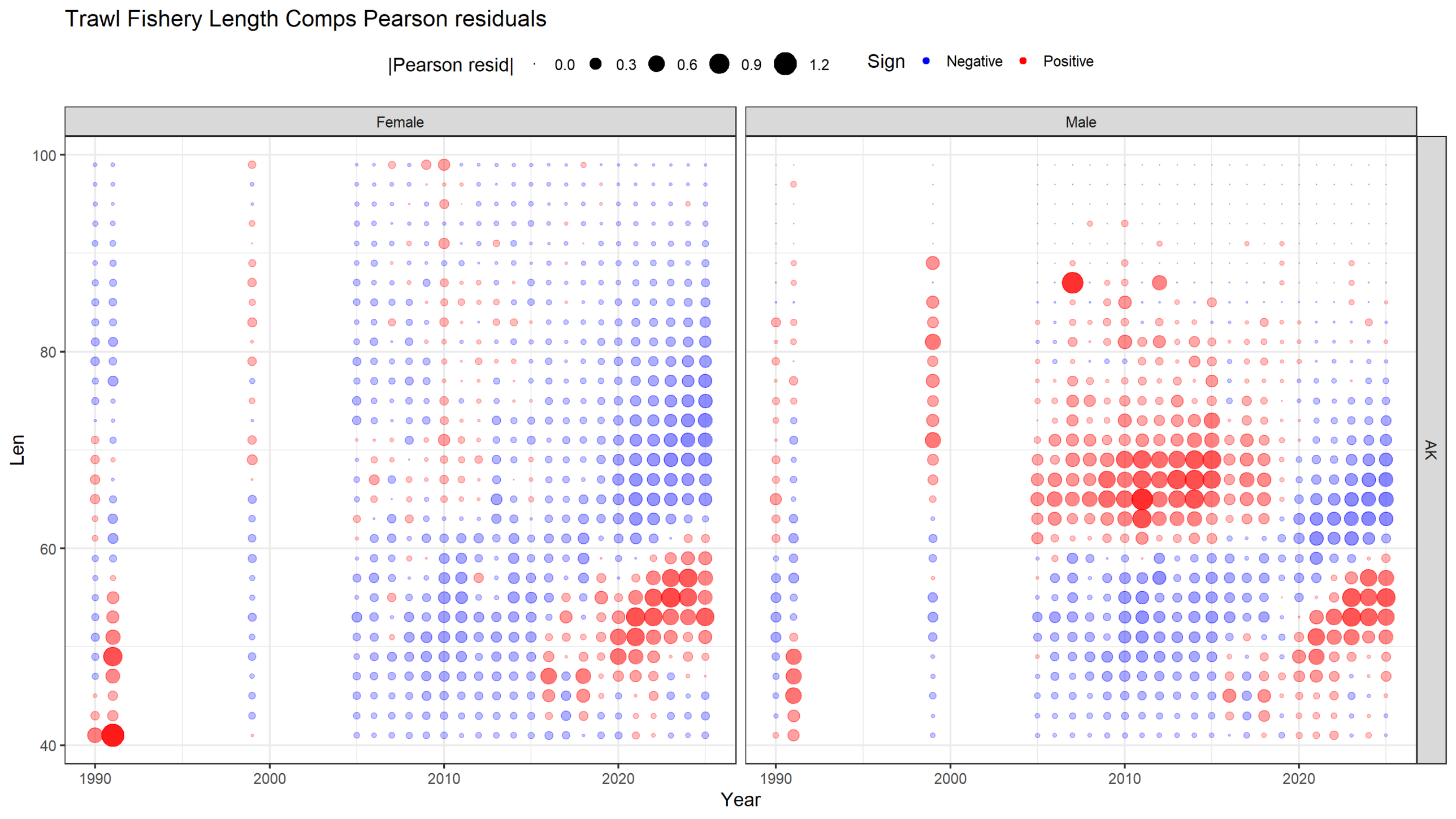
1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

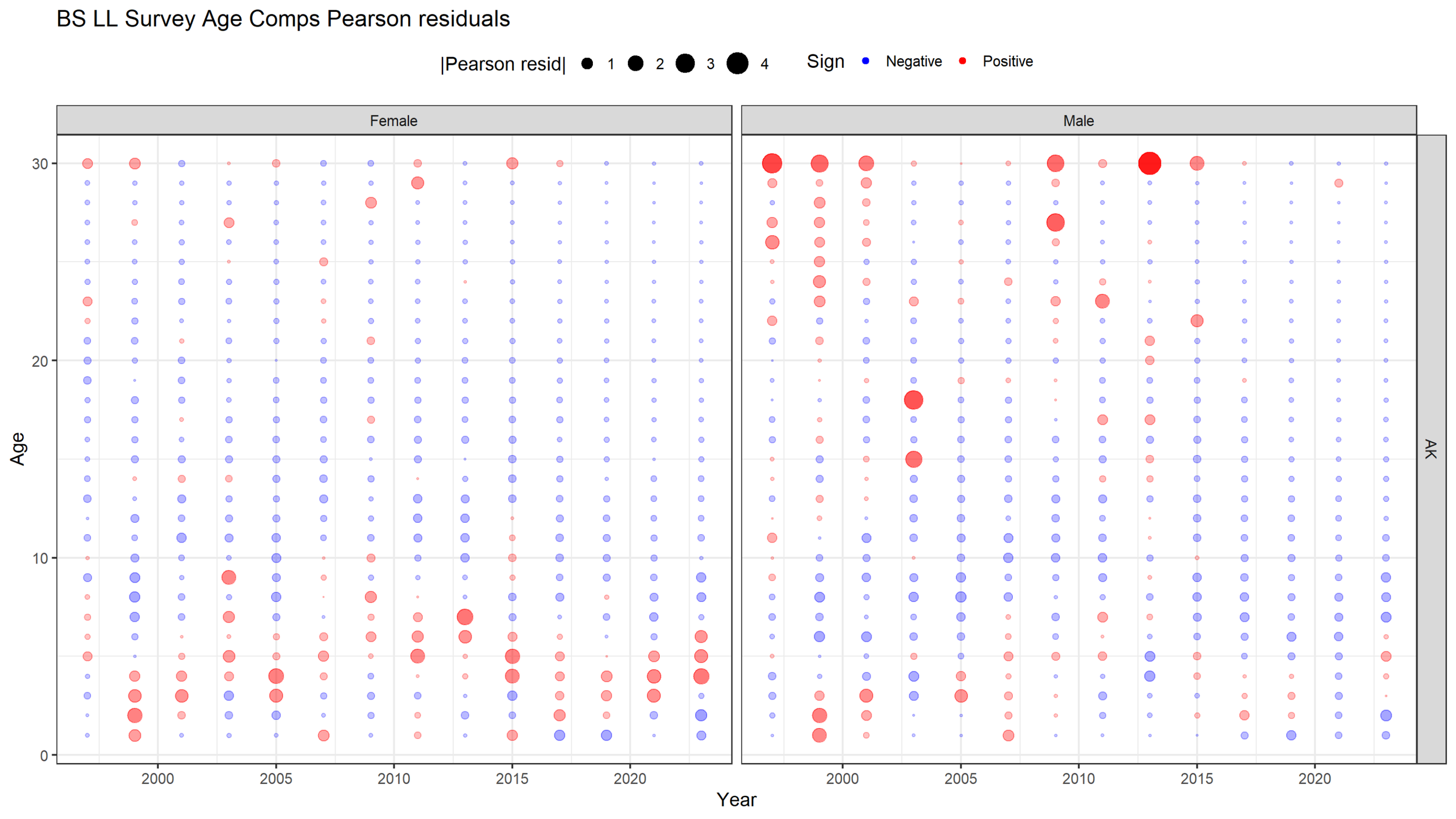
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

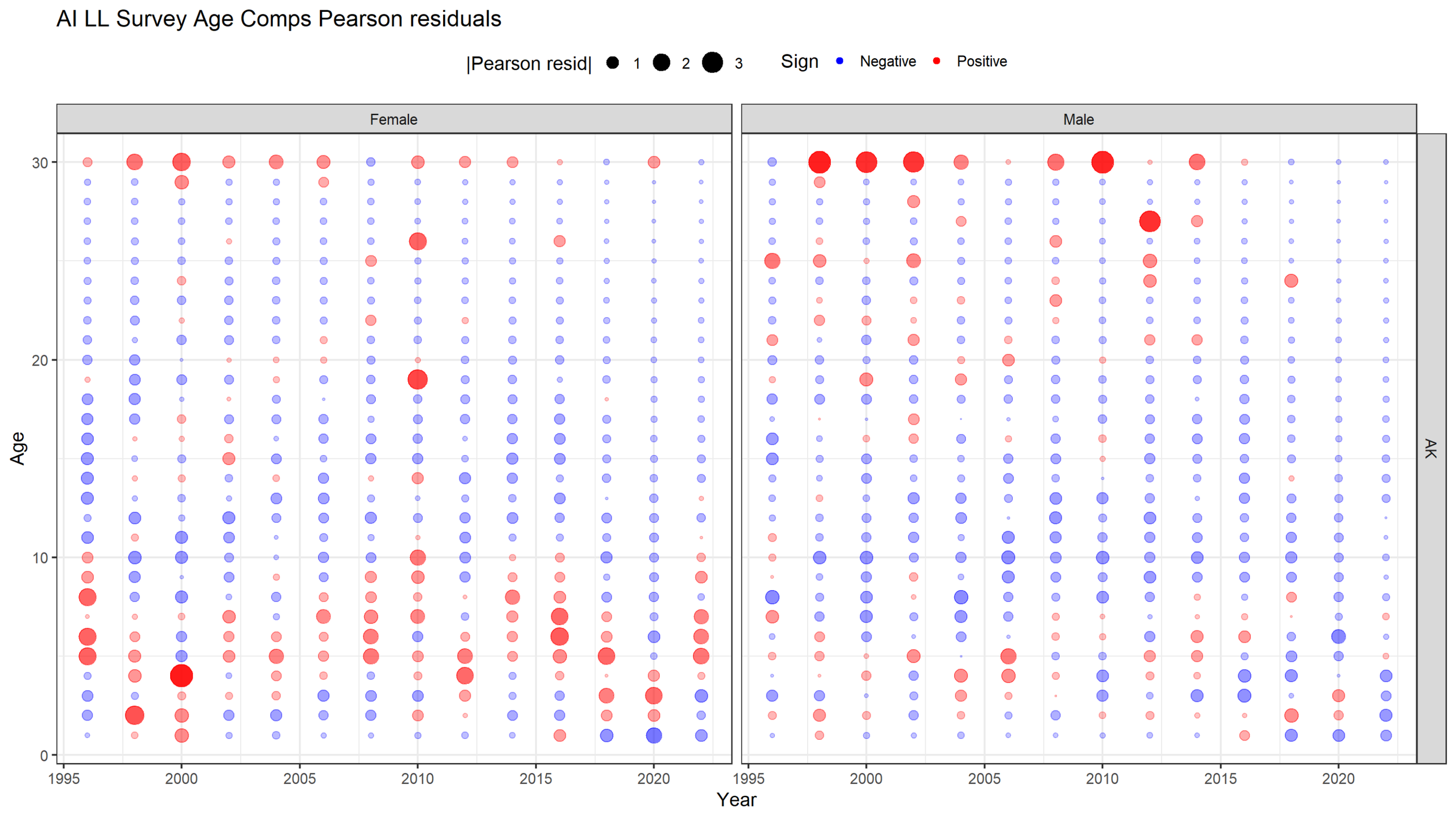
**JPN LL Survey Length Comps OSA AK**

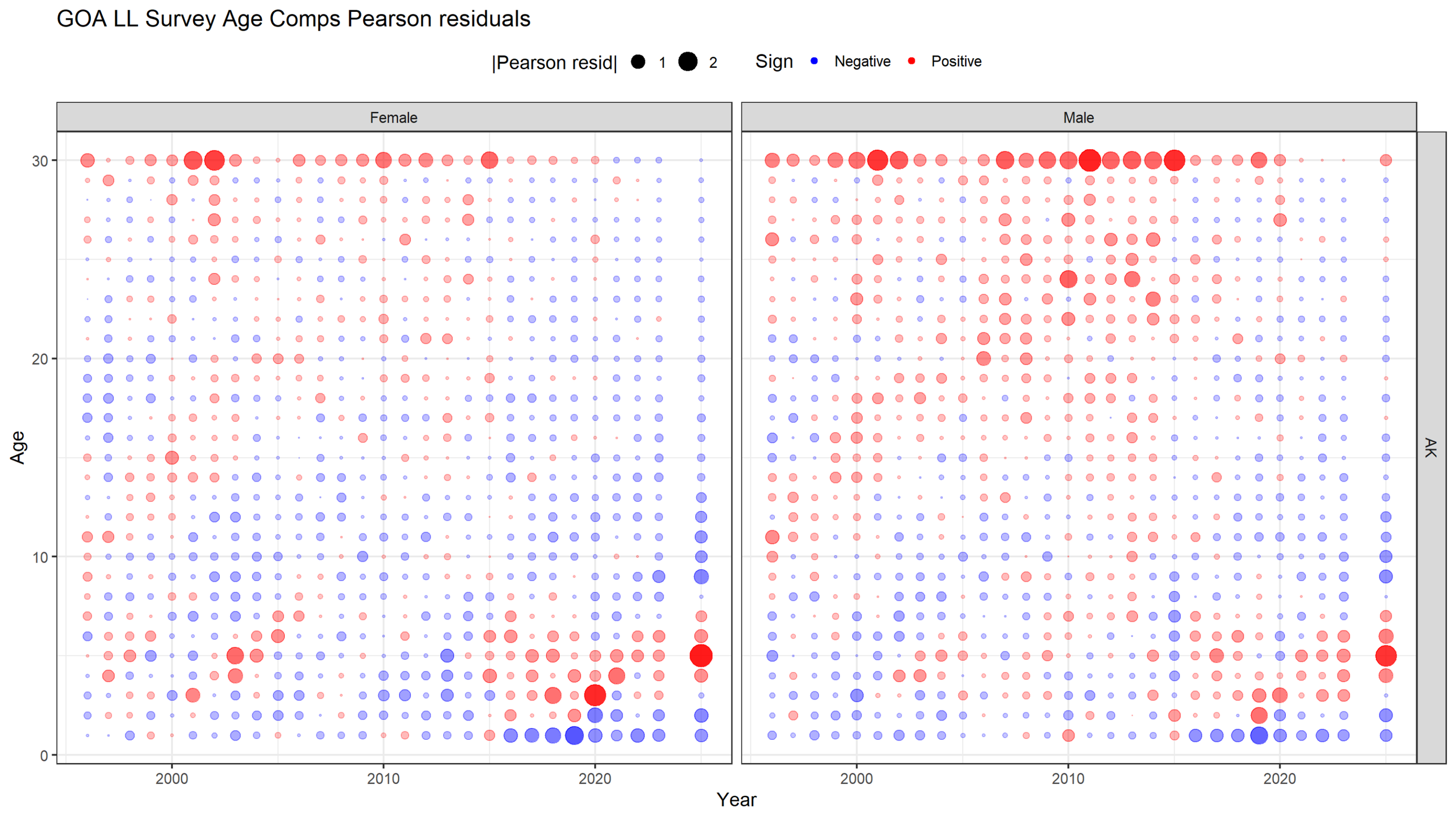


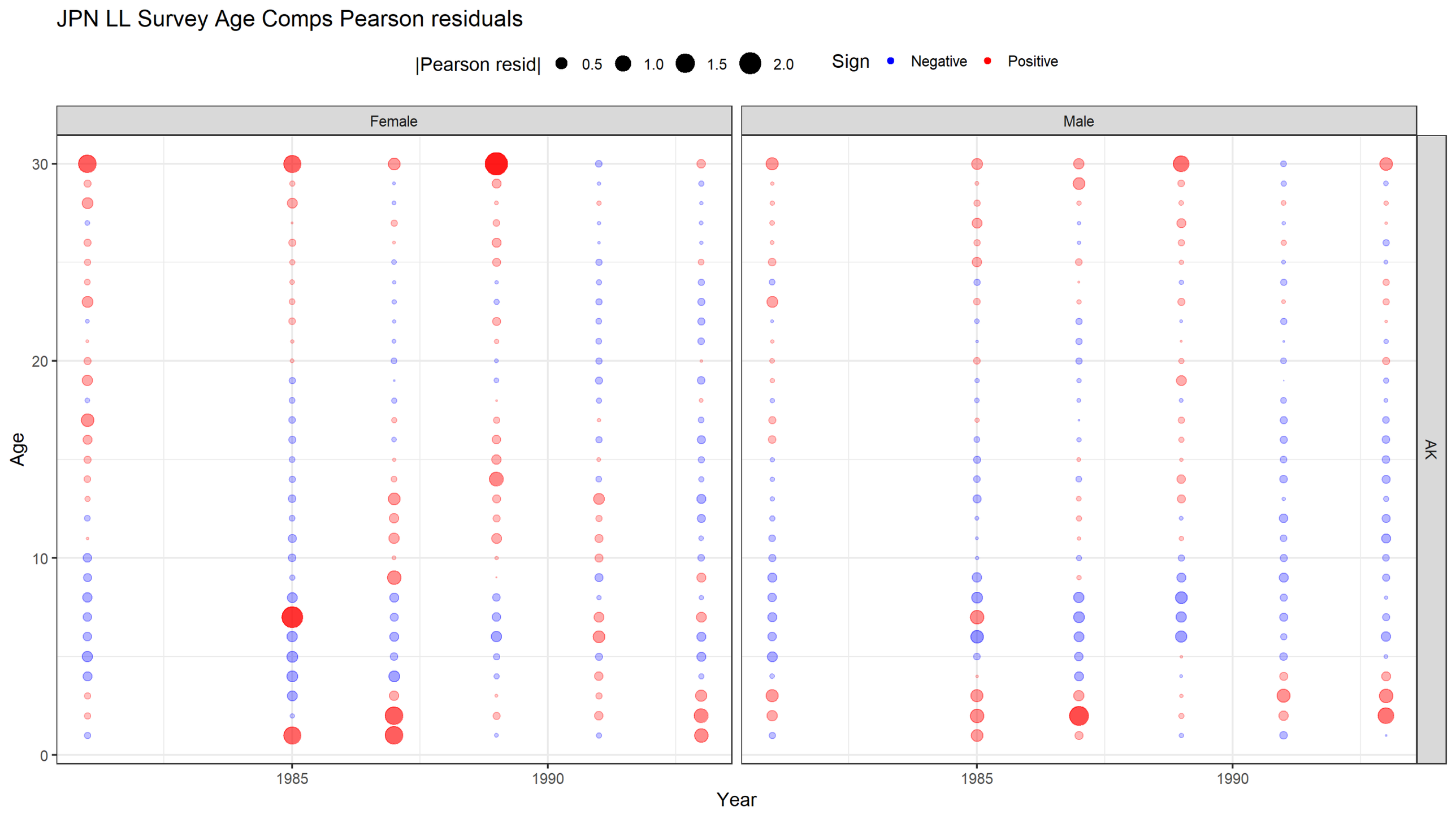

Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid| 1 2
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

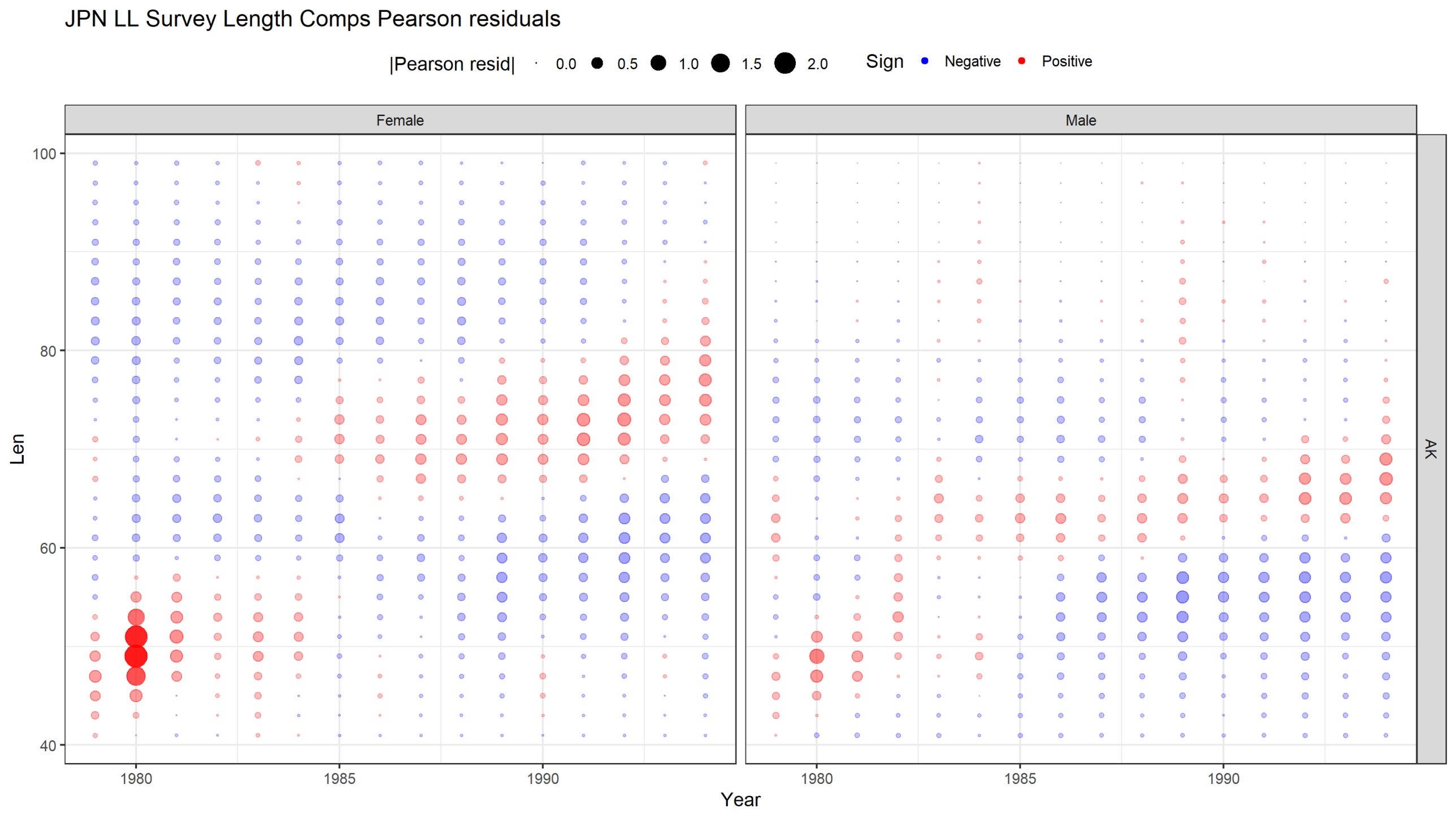

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.0
0.3
0.6
0.9
1.2
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year


BS LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3 4
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020


AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020


GOA LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020


JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year


JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

| Source | SE | Data_Wt | ISS | ESS |
|---|---|---|---|---|
| BS LL Survey Index | 0.20 | 1.00 | NA | NA |
| AI LL Survey Index | 0.23 | 1.00 | NA | NA |
| GOA LL Survey Index | 0.08 | 1.00 | NA | NA |
| Trawl Survey Index | NA | NA | NA | NA |
| JPN LL Survey Index | 0.07 | 1.00 | NA | NA |
| Fixed Gear Fishery CPUE | NA | NA | NA | NA |
| BS LL Survey Ages | NA | 0.61 | 34.07 | 20.93 |
| AI LL Survey Ages | NA | 2.00 | 35.57 | 71.11 |
| GOA LL Survey Ages | NA | 0.37 | 67.17 | 24.72 |
| JPN LL Survey Ages | NA | 0.15 | 83.33 | 12.10 |
| JPN LL Survey Lengths | NA | 0.34 | 54.62 | 18.48 |
| Fixed Gear Fishery Ages | NA | 0.04 | 142.85 | 6.22 |
| Trawl Fishery Lengths | NA | 0.09 | 47.62 | 4.41 |

| Model | Region | Terminal_SSB | Terminal_SSB0 | Terminal_F | Catch_Advice | B_Ref_Pt | F_Ref_Pt | B_over_B_Ref | B_over_DynB_Ref | F_over_F_Re |
|---|---|---|---|---|---|---|---|---|---|---|
| 26.2_FAA_M_Pr_0.115 | 1 | 264.1334 | 394.9323 | 0.0415361 | 57.02114 | 119.3839 | 0.09446 | 2.21247 | 0.66881 | 0.43973 |

Phase Plane Assuming B40 from ~2018

1995
1990
2000
2005
2010
1975
2015
1970
2020
1980
1985
2025
2026
1965
1960

1.0
0.5
0.0

$F/F_{35\%}$

0
1
2
3

$SSB/B_{35\%}$

Year
2020
2000
1980
1960

Control Rule
OFL (F35%)
Max ABC (F40%)